

Redox-Based Resistive Switching Memories – Nanoionic Mechanisms, Prospects, and Challenges

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1. Introduction

The ultimate nonvolatile data memory (NVM) should display characteristics such as high-density and low cost, fast write and read access, low energy operation, and high performance with respect to endurance (write cyclability) and retention.^[1] Today, Si-based Flash memory devices represent the most prominent NVM because of their high density and low fabrication costs. However, Flash suffers from low endurance, low write speed, and high voltages required for the write operations. In addition, further scaling, i.e., a continuation in increasing the density of Flash is expected to run into physical limits in the near future. Ferroelectric random access memory (FeRAM) and magnetoresistive random access memory (MRAM) cover niche markets for special applications. One reason among several others is that FeRAM as well as conventional MRAM exhibit technological and inherent problems in the scalability, i.e., in achieving the same density as Flash today. To overcome the problems of current NVM concepts, a variety of alternative memory concepts is explored. Most notably, NVMs based on electrically switchable resistance have attracted considerable attention, often summarized under the umbrella term resistance (switching) random access memory, short RRAM. This review will cover particularly interesting classes of RRAM in which redox reactions and nano-ionic transport processes play the key role. It should be noted, however, that despite the quite consistent pictures painted in this review many details and many variants are still completely unknown and our current pictures have more the character of working hypotheses instead of well funded physical models.

2. Classification of RRAM Types and Physical Storage Mechanisms

2.1. General Requirements of RRAM

Memory cells in a RAM are organized in a matrix. The rows and columns of the matrix are called word lines and bit lines,

respectively, connecting to the electronic amplifiers in the periphery of the matrix which conduct the write and read operations. In the simplest case, resistively switching memory cells may be organized in a passive cross-bar matrix, just connecting the word and bit lines at each node (Fig. 1a). In order to avoid the so-called parasitic-path-problem, i.e., signal bypasses by cells in their low resistance state, serial elements with a particular non-linearity must be added at each node. Depending on the switching scheme of the memory cell, these can be diodes or varistor-type elements with a specific degree of non-linearity.^[2] Alternatively, a RAM is organized in an *active* matrix comprising of a select transistor at each node which decouples the memory cell if it is not addressed (Fig. 1b). This concept significantly reduces crosstalk and disturbs signals in the matrix at the expense of some additional area required for the footprint of the transistor contacts.

A resistive switching memory cell in a RRAM is generally built by a capacitor-like MIM structure, composed of an insulating or resistive material 'I' sandwiched between two (possibly different) electron conductors 'M'. In the framework of this review, the material 'I' are oxides or higher chalcogenides which typically show some ion conductivity. These MIM cells can be electrically switched between at least two different resistance states, after an initial electroforming cycle which is usually required to activate the switching property. By applying appropriate programming or write voltage pulses V_{wb} , a cell in its high-resistance (OFF) state can be SET to a low-resistance (ON) state or RESET back into the OFF state. In the literature, the RESET is sometimes called an 'erase' operation. In a number of cases multilevel switching has been demonstrated, i.e., more than two resistance states have been established in order to realize, for example, multiple bits per

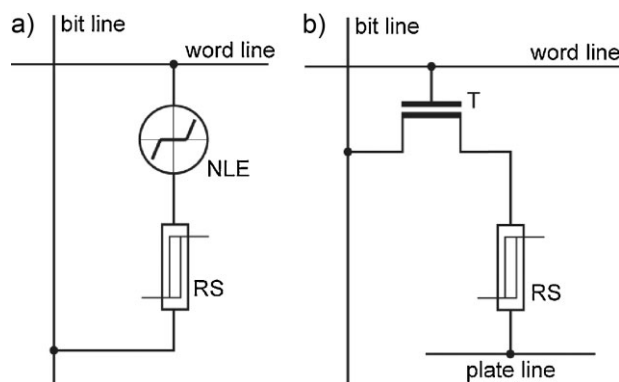


Figure 1. Circuit diagram of a storage node in the matrix of a resistance random access memory (RRAM), where RS denotes the resistive switching cell. a) Passive matrix, in which NLE is a serial element with a specific non-linearity. b) Active matrix with the select transistor T.

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cell. The state of the RRAM cell is detected by applying a read voltage V_{rd} .

Based on the circuit requirements of high-density NVM today such as Flash and taking predictions about technology scaling of the next 15 years into account,^[3] one can collect a number of requirements for RRAM cells:

Write operation

Write voltages V_{wr} should be in the range of a few hundred mV to be compatible with scaled CMOS to few V (to give an advantage over Flash which suffers from high programming voltages). The length of write voltage pulses t_{wr} is desired to be <100 ns in order to compete with DRAM specifications and to outperform Flash which has a programming speed of some 10 μ s, or even <10 ns to approach high-performance SRAM.

Read operation

Read voltages V_{rd} need to be significantly smaller than write voltages V_{wr} in order to prevent a change of the resistance during the read operation. Because of constraints by circuit design, V_{rd} cannot be less than approximately one tenth of V_{wr} . An additional requirement originates from the minimum read current I_{rd} . In the ON-state, I_{rd} should not be less than approximately 1 μ A to allow for a fast detection of the state by reasonably small sense amplifiers. The read time t_{rd} must be in the order of t_{wr} or preferably shorter.

Resistance ratio

Although an R_{OFF}/R_{ON} ratio of only 1.2 to 1.3 can be utilized by dedicated circuit design as shown in MRAM, R_{OFF}/R_{ON} ratios >10 are required to allow for small and highly efficient sense amplifiers and, hence, RRAM devices which are cost competitive with Flash.

Endurance

Contemporary Flash shows a maximum number of write cycles between 10^3 and 10^7 , depending on the type. RRAM should provide at least the same endurance, preferably a better one.

Retention

A data retention time t_{ret} of >10 years is required for universal NVM. This retention time must be kept at thermal stress up to 85 °C and small electrical stress such as a constant stream of V_{rd} pulses.

The combination of requirements on the write operation, the read operation, and the retention sets a voltage-time dilemma which is not addressed in most of the papers published on resistive switching so far. A ratio V_{wr}/V_{rd} of ten at most needs to lead to an acceleration of the switching kinetics of t_{ret}/t_{wr} , i.e., approx. 10^{16} ! There are only a few physical mechanisms which show such a huge non-linearity. We will address this voltage-time dilemma in Sections 3 to 6.



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It is interesting to note that resistance switching memory cells can be described as so-called memristors, which have been predicted by Leon Chua as the fourth basic circuit element because of the conceptional symmetry with the resistor, inductor, and capacitor,^[4] later extended to memristive devices.^[5] He described a generalized flux ϕ as time integral of voltage $V(t)$ and relates this to the charge q as time integral of current $I(t)$. If the resistance representing the ratio V/I depends on the charge $q(t)$ passed through the device, it becomes a memristance $M(q(t))$ according to

$$V(t) = M(q(t))/t \quad (1)$$

In this case, the memresistance shows hysteretic behavior which can be exploited as non-volatile resistance switching memory cells. The interesting link between the theoretical concept of Chua and the body of literature on resistance switching memory cells was first pointed out by Stanley Williams and his colleagues.^[6] This description is particularly useful when multi-level resistance values or even analogue values are to be stored and processed (cf. Sec. 6).

In order to describe the different resistive switching modes, we need to distinguish between two schemes with respect to the electrical polarity required for resistively switching cells. The switching operation is called unipolar when the switching procedure does not depend on the polarity of the write voltage signal (Fig. 2a). During the SET process, the current is typically limited by the current compliance of the control circuit. The RESET into the OFF-state occurs at a higher current and a voltage below the SET voltage. The switching operation is called bipolar when the SET to an ON-state occurs at one voltage polarity and the RESET to the OFF-state on the reversed voltage polarity (Fig. 2b). The MIM system needs to have some asymmetry, such as

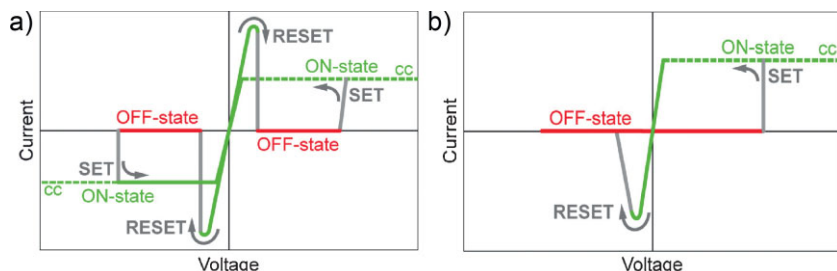


Figure 2. The two basic operation schemes of resistance switching memory cells. *I*–*V* curves recorded for a triangular shaped voltage signal. *cc* denotes the compliance current. Dashed lines indicate that the real voltage at the system will differ from the control voltage because of the *cc* in action. a) Unipolar switching. The SET voltage is always higher than the RESET voltage, and the RESET current is always higher than the *cc* during SET operation. b) Bipolar switching. The SET operation occurs on one polarity of the voltage or current, the RESET operation requires the opposite polarity. In some systems, no *cc* is used. Please note that the *I*–*V* curves of real systems may deviate considerably from these sketches, for both operation schemes.

different electrode materials ‘M’ or a dedicated voltage polarity during the initial electroforming step, in order to show bipolar switching behavior. Later in the text we will typically refer to an active electrode (AE) at which the bipolar switching takes place and an (ohmic) counter electrode (CE). It must be noted that the sketches in Figure 2 only show the principle *I*–*V* behavior with respect to the switching direction. Depending on the specific system, the curves may vary significantly. Systems which can be changed by the operation conditions between unipolar and bipolar operation have been named nonpolar.^[7]

2.2. Resistive Switching Mechanisms and the Role of Defects

A large variety of physical phenomena are known, which can – in principle – lead to non-volatile resistive switching memory effects (Fig. 3). The actual physical driving force of the resistance switching, although electrically induced in all cases, is quite different. Mechanical forces can be utilized in nanomechanical memory effects. A change in the molecular configuration may be developed into a resistance memory effect of a single molecule.^[8] Electrostatic and electronic effects are discussed as the possible origin of resistive switching (see Sec. 4.1). The direction of a ferroelectric or a ferromagnetic domain polarization may lead to different tunnel currents.^[9,10] The temperature induced change between a crystalline phase (ON-state) and an amorphous phase (OFF-state) in dedicated tellurides and selenides is exploited in phase change memories (PCM).^[11] And there are three classes which involve chemical effects, i.e., effects which relate to redox processes in the MIM cell either triggered by temperature or electrical voltage or both. These three RRAM classes are covered in the present review. Firstly, the bipolar electrochemical metallization mechanism or memory effect (ECM, Sec. 3) relies on an electrochemically active electrode metal such as Ag, the drift of the highly mobile Ag⁺ cations in the ion conducting ‘I’ layer, their discharge at the (inert) counterelectrode leading to a growth of Ag dendrites, which form a highly conductive filament in the ON state of the cell.^[12] Upon reversal of the polarity of the

applied voltage, an electrochemical dissolution of these filaments takes place, resetting the system into the OFF state. Secondly, the valence change mechanism or memory effect (VCM, Sec. 4) occurs in specific transition metal oxides and is triggered by a migration of anions, such as oxygen anions (which are typically described by the motion of the corresponding vacancies, i.e., oxygen vacancies). A subsequent change of the stoichiometry leads to a redox reaction expressed by a valence change of the cation sublattice and a change in the electronic conductivity. This bipolar memory switching is induced by voltage pulses, where the polarity of the pulse determines the direction of the change, i.e., reduction or oxidation. A third class relies on a thermochemical mechanism or memory effect (TCM, Sec. 5) which leads to a change of the stoichiometry due to a current-induced

increase of the temperature. The review will describe the major similarities and differences between these three classes, summarize the current state of knowledge, and outline future work in this area.

Of course, the grouping of the classes could be done in different ways. With emphasis on the material, one could comprise PCM, TCM, and VCM systems into one group in which oxides and higher chalcogenides play the dominant role as the insulating or resistive materials ‘I’ and the electrode materials play a – still relevant – secondary role. In contrast, the metal electrodes ‘M’ are crucial for ECM systems and for memories based on (pure) electrostatic/electronic effects while the material ‘I’ is of secondary influence in these systems. A grouping

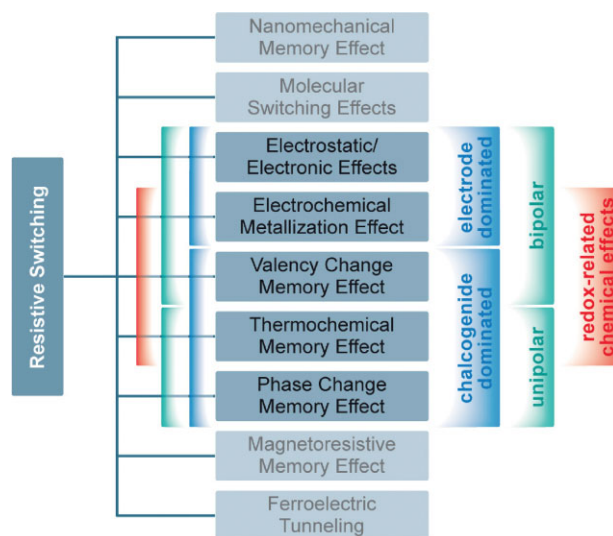


Figure 3. Classification of the resistive switching effects which are considered for non-volatile memory applications. The switching mechanisms based on thermal, chemical, and electronic/electrostatic effects (five classes in the center) are further described in the text. This review will cover the redox-related chemical switching effects (red bracket).

according to the switching schemes would show PCM and TCM in the group of the unipolar switching mode and the other classes in the group of the bipolar switching mode. It should be noted that a further grouping is conceivable in which the PCM effect is closely related to the three redox-based effects covered by this review. It concerns the key role of lattice defects which are essential for the microscopic switching mechanism in all these classes. The chalcogenides which show the PCM accommodate up to 25% of vacancies in the crystalline phase. During switching, the kinetics of the phase transformation is accelerated by several orders of magnitude due to the presence of these defects. Lattice defects play a similarly essential role in the redox-related memory effects as will be described in Sections 3 to 5.

Historically, the first papers on resistance switching in oxide insulators were published in the 1960s (although similar electrochemical effects involving valve metal oxides date back to even before). A huge variety of materials in an MIM configuration have been reported to show hysteretic resistance switching, comprehensively reviewed by Dearnaley et al.,^[13] Oxley,^[14] and Pagnia.^[15] Most of the papers cited in there describe unipolar switching effects, which we would consider as thermochemical memory switching (TCM) although often the experimental data are insufficient to make a definite assignment. A new era in the research activity started in the late 1990s, triggered by Asamitsu et al.,^[16] Kozicki et al.,^[12] and Beck et al.,^[17] recently reviewed by Waser and Aono^[18] and Sawa.^[19] The present review focuses on the microscopic understanding of the redox-based ion-migration related switching processes and the essential role which defects play in these switching phenomena.

3. Electrochemical Metallization Systems

3.1. Basic Principle of Operation

In electrochemical metallization (ECM) cells, also called Conductive Bridging (CB) cells or Programmable Metallization Cell (PMC) in the literature, an electrochemical metal deposition and dissolution is utilized to perform the resistive switching operation.^[20] Figure 4 schematically shows the basic principle of operation of an ECM memory cell in conjunction with an I - V switching cycle using a quasi-static triangular voltage V stimulation signal. The cell consists of an electrode made from an electrochemically active metal M , such as Ag, Cu, or Ni, an electrochemically inert counter electrode (CE), such as Pt, Ir, W, or Au, and a thin film of a solid electrolyte ('I'), i.e., a M^{z+} ion-conductor, sandwiched between both electrodes. In the initial high resistance state (OFF state) of the cell, no electrodeposit of the metal M is present on the inert electrode (Fig. 4D).

Hence, in the OFF state the cell has to be considered as an electrochemical half cell comprising of the active electrode (AE) and the adjacent electrolyte layer, a serial resistance representing the electrolyte film, and a serial capacitor comprising of the inactive counter electrode (CE) and the adjacent electrolyte layer (Figure 5). Please note that throughout the entire paper, if there is an active electrode it is shown in the figures on the left side of the resistive switching cell.

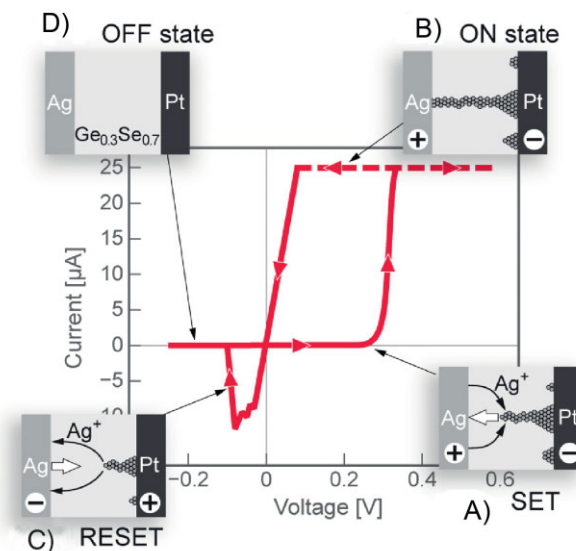


Figure 4. Typical current-voltage characteristic of a Ag/Ag-Ge-Se/Pt electrochemical metallization cell using a triangular voltage sweep. Adapted from [21]. The ON conductance is limited by a compliance current of 25 μ A. The curve in the compliance current state is shown as a dashed line because the actual voltage does not correspond to the voltage shown on the axis. The insets A to D show the different stages of the switching procedure (the phase-separated sub-structure of the Ag-Ge-Se solid electrolyte is not shown here for reasons of simplicity).

A SET process occurs (Fig. 4A) if a sufficiently positive bias voltage V is applied to the active electrode AE. The overall SET process involves the following steps:

- (i) anodic dissolution of M according to the reaction

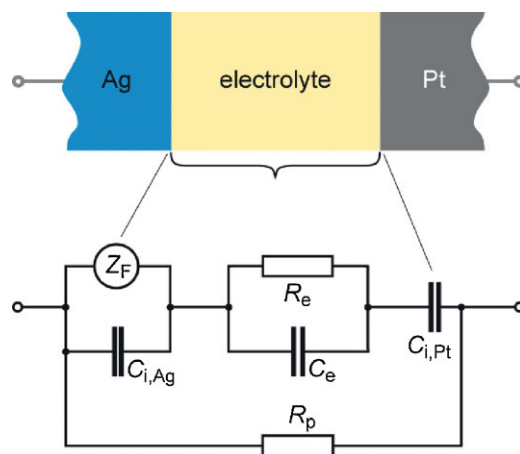


Figure 5. Equivalent circuit diagram of an ECM cell in the OFF state. Z_F denotes the Faraday impedance of the active electrode, $C_{i,Ag}$ and $C_{i,Pt}$ are the interface capacitances of the electrochemically active and inert electrode, respectively. R_e and C_e represent the resistance and capacitance of the electrolyte. R_p denotes a (typically very high) parallel resistance based on the electronic leakage current through the cell due to, for example, thermoionic emission and hopping conduction, or tunnelling in case of very thin electrolytes.

where M^{z+} represents the metal cations in the solid-electrolyte thin film;

- (ii) migration of the M^{z+} cations across the solid-electrolyte thin film under the action of the high electric field;
- (iii) reduction and electrocrystallization of M on the surface of the inert electrode CE according to the cathodic deposition reaction



The electrocrystallization process (Eq. 3) is electric field-enhanced and leads to formation of a metal filament growing preferentially in the direction of the active electrode (cf. Fig. 4A). After the metal filament has grown so far to make a galvanic metallic contact to the opposite M electrode, the cell has switched to the ON state (Fig. 4B). The cell retains the ON state unless a sufficient voltage of opposite polarity is applied and the electrochemical dissolution of the metal filament RESETs the cell (Fig. 4C) to its initial OFF state (Fig. 4D). Please note that in the initial phase of RESET there is an electronic current through the metallic bridge and, in parallel, an electrochemical current (also called Faradaic current) which dissolves the metal filament. The switching speed of ECM cells is determined mainly by the kinetics of the involved process step (i) to (iii). Therefore, in the next section the specific theoretical and experimental aspects of these process steps will be considered in more detail in order to track down the rate limiting step which finally controls the overall kinetics. The discussion will be on the basis of results obtained in selected ECM systems.

The most frequently studied ECM systems involve Ag and Cu as electrochemically active metals and phase separated amorphous selenides and sulfides as well as various oxides, acting as solid electrolytes.^[18] A resistive switching due to formation and dissolution of a silver dendrite was already reported in 1976 by Hirose and Hirose,^[22] who used Ag-doped amorphous As_2S_3 as solid electrolyte in a lateral system Ag/Ag- As_2S_3 /Au. Figure 6 reproduces an optical micrograph from their original paper showing the formation of a single Ag filament. Depending on the geometry of the cell and the experimental conditions it is conceivable that multifilament switching occurs as well. How-

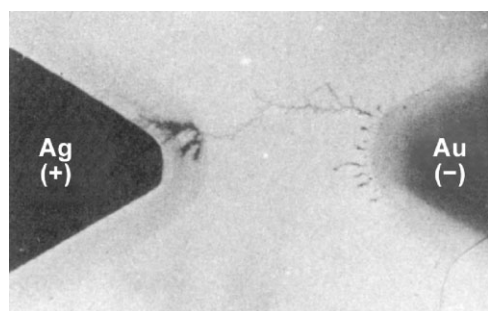


Figure 6. Optical microscopy image of a Ag dendrite grown from the (-)Au electrode towards to (+)Ag electrode within a As_2S_3 thin film on a glass substrate. Reproduced with permission from [22]. Copyright 1976, AIP.

ever, many reports published so far are consistent with single filament switching.

During the last decade, the development of ECM devices has been advanced, in particular, by Kozicki and colleagues^[12,20,23–25] as well as by Aono and colleagues.^[26–29] Predominantly, Ag- and Cu-doped amorphous Ge-Se,^[25–30] Ge-S,^[31] and (Zn,Cd)S^[32] films as well as oxide thin films such as SiO_2 have been applied as solid electrolytes. Switching times of less than 100 ns and resistance ratios R_{OFF}/R_{ON} of more than five orders of magnitude were reported. Additional details about the electrolytes and their impact on the ECM effect are described in Section 3.3.

3.2. Cation Transport and Electrode Reactions

In some solid state ECM cells the amorphous thin film acting as matrix for the fast ion transport contains no cations M^{z+} of the electrochemically active metal after cell fabrication. This is one of the reasons why a forming procedure is typically required before the reproducible resistive switching can be utilized. In this case, the electroforming step is required to incorporate mobile M^{z+} ions in the insulating amorphous film matrix. Another reason may be the nano-morphological change in the electrolyte by the formation of a first metallic filament which obviously pre-forms the growth channel and may act as an easy transport channel for all subsequent switching events. A typical example represents the system Cu/ SiO_2 /Pt where the Cu-doping of the amorphous SiO_2 film can be achieved by thermal diffusion^[33] or by an electroforming process during the first current-voltage cycle as illustrated in Figure 7.^[34]

As seen in Figure 7a, at constant voltage sweep rate, the voltage $V_{SET,form}$ to start the SET process during the electroforming cycle is significantly higher than the SET voltage V_{SET} for the subsequent switching cycles. Furthermore, $V_{SET,form}$ increases linearly with the thickness of the SiO_2 -film, whereas the SET voltage for all subsequent switching events, V_{SET} , is almost thickness independent (Fig. 7b). This behavior reveals that during the electroforming cycle the rate of the SET process may be determined by an electric field driven migration of Cu cations in the SiO_2 -film. After the formation of the first Cu filament and its dissolution during the first RESET, the remaining structural template of the dissolved filament may act as a fast transport and growth path. Nevertheless, the much lower SET voltage V_{SET} and its thickness independence observed during the subsequent switching cycles can be explained by a path for fast ion transport and preferential filament growth that is formed in the amorphous SiO_2 -film during the first electroforming cycle.

As already discussed in the previous section, the resistive switching in the ECM memory systems includes transport of metal cations M^{z+} in the thin electrolyte film, step (ii), and electrode reactions involving transfer of M^{z+} across both electrode/electrolyte interfaces, steps (i) and (iii), and electrocrystallization phenomena, step (iii). We will first discuss the I - V dependencies of the three sequential steps (i) to (iii) in general. Subsequently, we will apply the equations to the specific experiment and use additional information in order to identify the possible rate-limiting step of the SET process.

The high mobility of metal cations M^{z+} in amorphous selenides, sulfides and oxides used in ECM systems is due to the

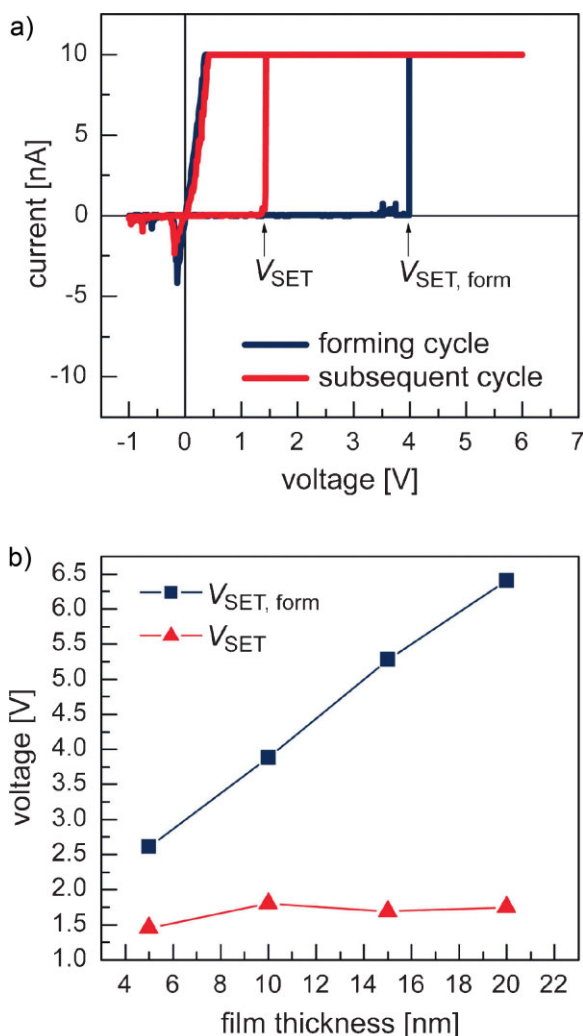


Figure 7. Current–voltage characteristic of a Cu/SiO₂ electrochemical metallization cell showing the first (forming) and subsequent cycles. (a) Forming and subsequent switching cycles measured at a sweep rate 1 V s^{−1} on a Cu/SiO₂/Pt cell with a 15 nm thick oxide layer and a circular 75 μm Cu top electrode. b) Oxide film thickness dependence of the SET voltage for the forming, V_{SET, form}, and subsequent switching cycles, V_{SET}.

long-range disorder in these materials and resulting fast transport paths. Assuming that M^{z+} is the only mobile ionic species, the ion transport, step (ii), will occur exclusively by migration of these cations in the fixed matrix of the amorphous material and can be described by the model of Mott and Gurney for an electric field driven, thermally activated ion hopping.^[35,36] The driving force of the ion migration is the electric field strength E , which is related to the potential drop $\Delta\varphi_{SE}$ across the ion-conducting film and the thickness d of the film by $E = \Delta\varphi_{SE}/d$. According to the hopping model the ion migration involves a series of thermally activated jumps between adjacent sites and in the case of a symmetric energy barrier W_a^0 the dependence of the ionic current density i on E is given by:^[36]

$$i = 2zecav \exp\left(-\frac{W_a^0}{kT}\right) \sinh\left(\frac{azeE}{2kT}\right) \quad (4)$$

where c is the concentration of mobile cations M^{z+}, a is the jump distance of the ions and ν is a frequency factor. For high electric fields ($E \gg kT/aze$), Equation 4 predicts an exponential dependence of i on E

$$i = zecav \exp\left(-\frac{W_a^0}{kT}\right) \exp\left(\frac{azeE}{2kT}\right) \quad (4a)$$

whereas for low electric fields ($E \ll kT/aze$) it follows an Ohm's law type linear dependence of i on E

$$i = \frac{(ze)^2 c E}{kT} a^2 \nu \exp\left(-\frac{W_a^0}{kT}\right) = \frac{(ze)^2 c D \Delta\varphi_{SE}}{kTd} \quad (4b)$$

where $D = a^2 \nu \exp(-W_a^0/kT)$ represents the diffusion coefficient. Equation 4b corresponds to the Nernst–Einstein relation giving the temperature dependence of the ionic conductivity, which allows for a determination of the activation energy W_a^0 for ion hopping in absence of an electric field.

The current density for the charge transfer across the electrode/electrolyte interfaces during the cathodic reduction step (iii) leading to the metal deposition and the counter reaction representing the anodic oxidation and dissolution of M in the ECM cells, step (i), can be described by the Butler–Volmer equation:^[37]

$$i = i_0 \left[\exp\left(\frac{\alpha ze\eta}{kT}\right) - \exp\left(-\frac{(1-\alpha)ze\eta}{kT}\right) \right] \quad (5)$$

where i_0 is the exchange current density, α is the cathodic charge transfer coefficient and $\eta = \varphi_{eq} - \varphi$ represents the electrochemical overpotential defined as a difference between the equilibrium Nernst-potential φ_{eq} of the metal M and the actual electrode potential φ . A schematic representation of the charge transfer process described by Equation 5 is shown in Figure 8.

Note that in Equation 5 both, the cathodic overpotential and the cathodic current density are defined as positive quantities. For

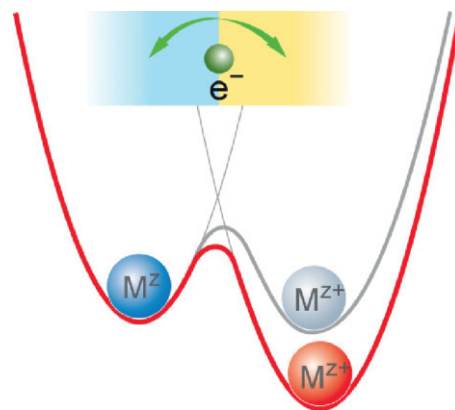


Figure 8. Sketch of the energy diagram of a charge transfer reaction at the interface between a metal cation at the surface of the metal electrode and a corresponding cation within the electrolyte as described by the Butler–Volmer equation. The grey line represents the situation with an overpotential η applied.

high cathodic overpotentials ($\eta \gg kT/ze$) Equation 5 transforms to

$$\ln i = \frac{\alpha ze}{kT} \eta + \ln i_0 \quad (5a)$$

This logarithmic relationship between i and η is known as Tafel equation and allows for an experimental determination of the charge transfer coefficient α and exchange current density i_0 .

Besides the charge transfer, step (iii) also includes the nucleation and growth of the metal phase. The electrodeposition current density i is related to the normal growth rate R of the metal phase by the Faraday law $i = zeR/V_M$, where V_M is the atomic volume of the metal.^[38] Considering the one-dimensional growth of a cylindrical metal filament in an ECM memory cell, the time t_{SET} for bridging both electrodes can be expressed by

$$t_{SET} = \frac{zed\pi r_f^2}{V_M I_f} \quad (6)$$

where I_f denotes the growth current and r_f the radius of growing metal filament, respectively.

Equation 6 accounts only for the growth of the metal filament during the SET process. However, in the ECM cells the electrocrystallization of M on the inert (foreign) substrate S starts with nucleation of the new metal phase.^[38,39] The electrochemical supersaturation $\Delta\tilde{\mu}$, which is the driving force for nucleation is related to the cathodic overpotential η by $\Delta\tilde{\mu} = ze\eta$. Due to its stochastic character, nucleation is a random phenomenon and the nucleation process can be considered as a sequence of random nucleation events characterized by a mean nucleation time $\bar{\tau}_n$, which is related to the stationary nucleation rate J ($\text{cm}^{-2} \text{s}^{-1}$) and the inert electrode area A by $\bar{\tau}_n = 1/JA$. This relation shows that the nucleation time will increase with decreasing inert electrode size. This behavior was applied successfully to the investigation of electrochemical nucleation and growth of single crystallites by using microelectrodes as substrates.^[38,40] As an example, Figure 9 shows as a single TI-crystallite formed by electrocrystallization on a Ag(111)-microelectrode.^[38]

A characteristic feature of nucleation is the existence of an overpotential threshold called critical overpotential η_{crit} , below which the nucleation rate is practically zero and above which it increases exponentially.^[38,39] The value of this critical overvoltage depends on the inert electrode material, the time scale of observation and on the measurement technique used to detect nucleation. For high cathodic overpotentials the critical nuclei usually consist only of several atoms and according to atomistic theory of nucleation the overpotential dependence of stationary nucleation rate can be expressed by the equation:^[38,39]

$$J = K(Z_0, N_c) \exp \left[\frac{(N_c + \alpha)ze}{kT} \eta \right] \quad (7)$$

where the pre-exponential term $K(Z_0, N_c)$ depends on the number density Z_0 of the nucleation sites and on the number N_c of atoms constituting the critical nucleus.

Information on the kinetics of the SET process in the ECM cells can be obtained by potentiostatic current transient measurements

(also called voltage step experiment) or by potentiodynamic I - V measurements at different voltage sweep rates $\nu = dV/dt$. Figure 10 shows a VSET versus $\log \nu$ plot of experimental data obtained from potentiodynamic I - V measurements in the ECM memory system Cu/SiO₂/Ir.^[41]

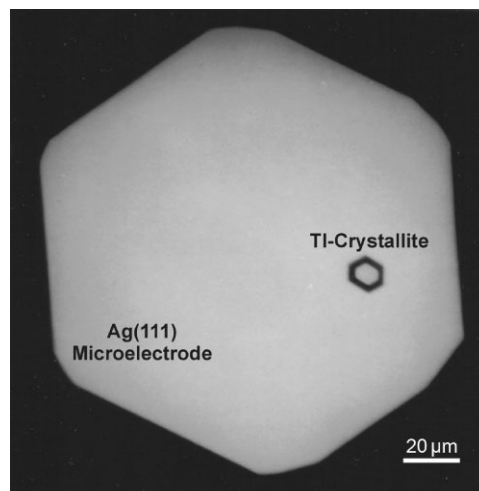


Figure 9. Example of an electrochemical nucleation and growth of single crystallites by using microelectrodes as substrates: a single TI-crystallite has formed by electrocrystallization on an Ag(111)-microelectrode. Reproduced with permission from [38]. Copyright 1996, Wiley-VCH.

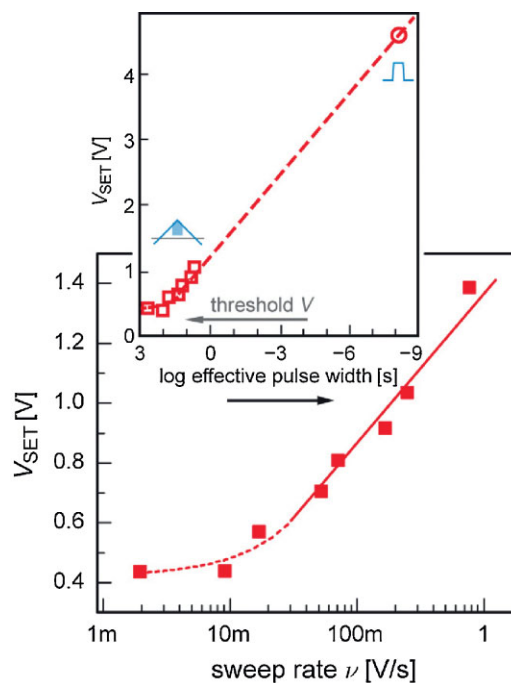


Figure 10. Switching voltage, V_{SET} , versus sweep rate, ν , measured on a Cu/SiO₂/Ir cell with an oxide thickness of 15 nm. For medium to high sweep rate ν , a clear exponential relationship between the switching voltage and the sweep rate is observed while for low ν -values a critical SET voltage is approached [41]. The inset puts the data into relation with a pulse measurement (dot) using a pulse width of 10 ns. The sweep rates of the triangular sweep experiments are converted into effective pulse width defined by one quarter of a full period.

A clear exponential relationship between the switching voltage and the sweep rate is observed for medium to high sweep rates ν while for low ν values a critical SET voltage is approached. The pronounced exponential relationship and, in particular, a critical threshold voltage for the SET process explain, how the voltage–time dilemma is overcome for the SET process in ECM cells. The conceivable microscopic origin of this exponential relationship is discussed in the following paragraph.

As shown above, all three steps of the SET process, (i) to (iii), described in Section 3.1 can exhibit an exponential relationship between the switching rate and the applied voltage. However, the anodic dissolution (i) at the Cu electrode will always be fast, because no crystallization overpotential is involved and also no concentration overpotential builds up due to the high field to which the Cu ions are exposed in the thin SiO₂ films. Step (ii), the ion migration process through the SiO₂ film is an electric field-driven, thermally activated hopping of Cu ions. Under extremely high electrical fields, ion migration may, in principle, become non-linear according to Equation 4a. However, an estimation of the hopping distance a based on the parameters given in the experiment corresponding to Figure 10 and assuming activation energies in the order of 0.5 eV to 1 eV leads to jump distance values a in the order of many nanometers instead of inter-atomic distances. Similarly large values of a were also reported for other systems.^[36] Because such large values of a , however, show no real physical meaning which renders the possibility of non-linear ion migration less likely. This conclusion is further supported by the fact that the SET voltage, V_{SET} , is almost not thickness dependent as shown in Figure 7b. The discussion about the potential impact of an accelerated ion conductivity at very high fields will be resumed in Section 4.6.

The above discussion leaves step (iii), the electrocrystallization of Cu at the inert Ir cathode, as the most probable rate-limiting step determining the kinetics of the SET process in the system Cu/SiO₂/Ir. In this case the applied voltage corresponds to the cathodic overpotential ($V \equiv \eta$). The electrocrystallization involves the initial nucleation of Cu on the Ir electrode and the subsequent Cu filament growth. If the latter process is rate-determining the growth current density can be described by the Butler–Volmer equation (Eq. 5b) for the cathodic charge transfer.

For the case of potentiodynamic electrodeposition by a linear voltage sweep with a sweep rate $\nu = dV/dt$ the charge Q_{SET} needed for the 1D growth of the Cu filament up to the on-switching can be expressed by:^[41]

$$Q_{\text{SET}} = \int_0^{V_{\text{SET}}} \frac{I_f}{\nu} dV = \int_0^{V_{\text{SET}}} \frac{i \pi r_f^2}{\nu} dV \quad (8)$$

where V_{SET} is the on-switching voltage, I_f is the growth current of the Cu filament, i is the electrodeposition current density. Using Equation 5a for i in Equation 8, one obtains the following dependence of the switching voltage V_{SET} on the voltage sweep rate ν

$$V_{\text{SET}} = \frac{kT}{\alpha ze} \ln \nu + \frac{kT}{\alpha ze} \ln \frac{Q_{\text{SET}} \alpha ze}{i_0 \pi r_f^2 kT} \quad (9)$$

As shown in Figure 10, current-voltage measurements with variable voltage sweep rates revealed that the switching voltage V_{SET} was strongly dependent on the sweep rate ν as predicted by Equation 9. Values $\alpha = 0.077$ and $\alpha = 0.154$ were estimated for the charge transfer coefficient from the slope of the solid line in Figure 10 and Equation 9 with $z = 2$ and $z = 1$, respectively. These values are smaller than those usually obtained in electrochemical systems with liquid electrolytes^[37] but are comparable to the value reported for a Ag-Ge-Se based solid state ECM cell.^[30] The logarithmic dependence of the switching voltage V_{SET} on the sweep rate ν indicates that the electrode kinetics in Cu/SiO₂ memory cells can be described by the Butler–Volmer equation. Equation 9 suggests that the switching voltage V_{SET} should also be dependent on the SiO₂-film thickness, because longer filaments have to grow in thicker films. Longer filaments require more Cu atoms, and therefore, more charge Q_{SET} has to be supplied. However, a nearly constant switching voltage was detected for ECM cells with the film thicknesses varying from 5 nm to 20 nm, as shown in Figure 7b. Assuming a filament diameter of 5 nm,^[42] the switching voltage according to Equation 9 would vary by ≈ 280 mV (for SiO₂-film thicknesses between 5 and 20 nm), which lies within the measurement variations. Due to these variations, the dependence of the switching voltage on the oxide thickness was probably not observed.

If the filament growth is relatively fast, the switching time $t_{\text{SET}} = V_{\text{SET}}/\nu$ can be identified with the time τ_1 required for the formation of the first Cu nucleus. In the case of stationary nucleation the mean value of this time corresponds to the mean nucleation time ($\bar{\tau}_1 \equiv \bar{\tau}_n$) and as shown above is inversely proportional to the probability of nucleation and consequently to the nucleation rate given by Equation 7. Therefore, based on the experimental data in Figure 10, the contributions of voltage dependences of the growth rate and of the nucleation rate cannot be separated, which leave this issue open to future studies. The observed asymptotic approach of a finite SET voltage at low sweep rates indicates a threshold voltage of approx. 0.4 V which may be caused by a nucleation overpotential of Cu on the inert Ir cathode. Further investigations are necessary in order to clarify the contribution of nucleation statistics to the kinetics of the overall SET process.

Considering the RESET operation, the question arises where the bridging metal filament connecting the two electrodes will start to dissolve. Because the active metal is present at both electrodes and forms the bridge, the answer to this question is not obvious. The question is: why would a chemically symmetrical cell (here, with Ag at both sides and forming the bridge) show a clear bipolar (i.e., antisymmetrical) switching? Therefore, experiments in a model Ag/H₂O/Pt ECM system with liquid aqueous electrolytes using coplanar electrode structures have been performed.^[43] This approach allows for an observation of growth and dissolution processes at different stages of the switching process by microscopy such as SEM. The observations in the system Ag/H₂O/Pt confirm the basic principle of operation of ECM cells. During the RESET operation, negative voltage polarity is applied to the Ag electrode and the positive polarity is applied to the (partially Ag coated) Pt electrode. Besides an electronic current across the Ag bridge, there is an electrochemical current through the electrolyte which is supported by the electrochemical dissolution of Ag at sites of positive potential

and Ag deposition at sites of negative potential at the surface of the metal. A numerical field simulation for this system based on the electrostatic potential at the metal surface revealed sites of electrochemical dissolution and sites of depositions. In agreement with experimental observations the simulation results show that the RESET process starts by Ag dissolution at the connective neck formed between the Ag dendrite and the planar counter Ir electrode, i.e., at the position at which the Ag filament first touched the Ir electrode during the SET process (Fig. 11).

3.3. Variety of Material Systems

As mentioned, the ECM effect is quite universal with respect to the type of electrolyte as long as the electrolyte exhibits a reasonably large conductivity of the cations of interest. Besides the “classical” binary sulfides and selenides and the ternary glassy

sulfides and selenides, there are various oxides such as WO_3 ,^[44] SiO_2 ,^[33,34] ZrO_2 ,^[45] and Al_2O_3 ,^[46] amorphous Si and C and ion-conducting polymers. From the broad range of electrolytes studied so far (see [20] for a survey) one can conclude that either the crystal lattice of the electrolyte allows for a fast diffusion of the cation of interest, e.g., Ag^+ in RbAgI_4 , or – much more frequent – the electrolyte shows an amorphous or defective crystalline structure in which fast transport channels are present.

As mentioned in Section 3.1, ternary glasses of higher chalcogenides represent a special case which exhibit specific advantages for ECM application. They are formed by dissolving the electrochemically active metal, Ag or Cu, in sulfides, selenides, or tellurides of Ge, As, Sb, or Ga. The dissolution process can be conducted by photo-induced and temperature assisted diffusion. The added Ag or Cu react with the chalcogen within the compound to form mobile cations. The mobility of the cations is quite high (approx. $10^{-10} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ at 298 K for Ag-Ge-S with 1 at % Ag^[47]) and the activation energies are low because of the relatively open structure of the amorphous chalcogenides, in which the cations easily find fast transport channels. Above a threshold concentration which is, for example, approx. 2 at % Ag in Ag-Ge-S, a phase separation occurs leading to crystalline metal chalcogenide (e.g., Ag_2S) nanoparticles dispersed in the amorphous matrix of the residual material. For example, if 40 at % Ag are added to $\text{Ge}_{0.30}\text{Se}_{0.70}$ then a dispersion of 20 at % Ag_2Se in $\text{Ge}_{0.30}\text{Se}_{0.50}$ results. The Ag_2Se nanoparticles show diameters in the range from 5 to 8 nm.^[25,48] Although the Ag_2Se constitutes a mixed electronically-ionically conducting phase, the concentration of Ag_2Se is below the percolation limit and, hence, a ECM cell with this electrolyte would be sufficiently high resistive in the OFF state. The advantage of such a system is revealed by the SET process because the Ag^+ ions only need to bridge the (small) gaps between the mixed-conducting nanoparticles in a coordinated process (Fig. 12). As a consequence, the SET process is considerably faster and less energy consuming compared to the case without the nanodispersed phase in which the Ag filament has to grow the entire distance from the cathode to the anode. The Ag randomly dissolved in the Ge-Se matrix is removed during the RESET process.^[49] In order to achieve very low leakage currents as a prerequisite for low write currents (e.g., 1 nA and below), a thin GeO_2 diffusion barrier has been introduced in the Ag-Ge-Se layer.^[21]

Besides these well studied chalcogenide glass systems and the oxide systems described before, there are numerous material systems which do not immediately look like ECM cells. However, in the experience of the authors it seems to be crucial to check if

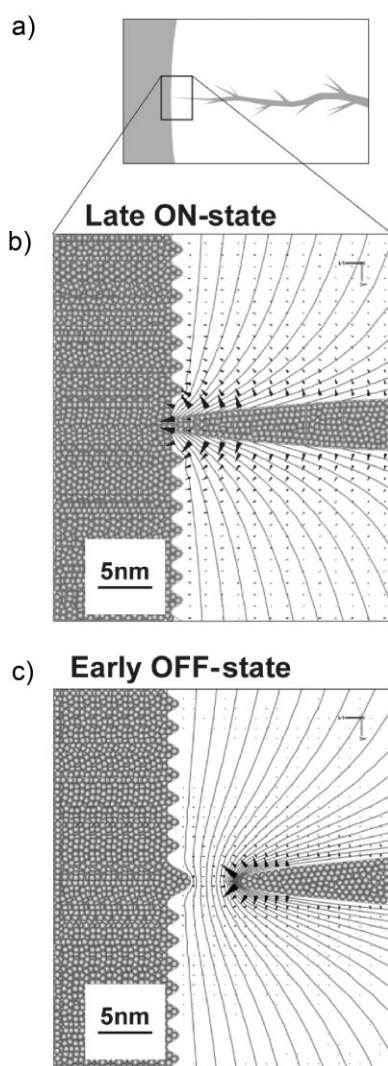


Figure 11. Field simulation of the front-most Ag filament touching the Ag electrode shown in a cross-section during the RESET process. The blue lines are equipotential lines. The black triangles show the direction and strength of the electrical field. a) sketch of the location, b) late ON state, c) early OFF state. Re-drawn with modifications from [43].

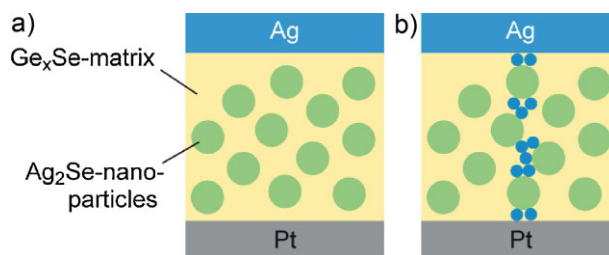


Figure 12. Sketch of an Ag/Ag-Ge-Se/Pt cell with nanodispersed Ag_2Se particles in a Ge-Se matrix. a) OFF state; b) ON state.

the ECM mechanism prevails whenever electrochemically active metals such as Cu or Ag are present in a cell showing resistive switching. For example, pronounced resistive switching was reported thirty years ago for Cu/Cu-TCNQ/Al cells where TCNQ denotes tetracyanoquinodimethane.^[50] In the original work, a charge transfer reaction between Cu and TCNQ was claimed to be responsible for the resistive switching effect. A variety of control experiments and additional experimental techniques recently revealed that the resistive switching in fact is based on a ECM mechanism in which a thin Al oxide/Al hydroxide layer acts as the electrolyte through which a Cu filament grows.^[46,51] The oxide/hydroxide layer is inevitably formed during the preparation of the cells in the sequence described in the literature.

4. Valence Change Systems

4.1. General Features

Many binary transition metal oxides as well as multinary oxides with at least one transition metal sublattice show bipolar resistive switching even though the electrodes do not inject metal cations. This missing cation injection may be either because the electrode metal is not easily oxidized (in the case of Pt, Au, etc.) or the oxidized form is not easily reduced back to the metal (in the case of Al, Ti, Nb, etc.).

First reports which seem to fall into this category go back to the 1960s when Nb/NbO_x thin film cells were studied.^[52] In the late 1990s, Tokura, Kawasaki and their colleagues started to study bipolar resistive switching phenomena for various manganites^[16,53] while Bednorz and his colleagues focused on titanates and zirconates.^[17,54] As in the case of the ECM cells, usually an electroforming step is required before bistable switching is achieved. Often, the electroforming is a somewhat slower process (milliseconds to thousands of seconds), than the actual switching depending on the geometry of the system (thin film MIM structures, lateral MIM cells, single crystals) and the electroforming parameters. The polarity of the bipolar switching cycle is not always obvious. It seems to be determined by many factors such as the work function and the oxygen affinity of the electrode metals, in particular, if different metals are used for the two electrodes of a cell, the polarity of the electroforming process, the formation of interface layers, etc. Throughout the following text, some of the parameters mentioned will be explained.

In the literature, a broad spectrum of electronic and/or ionic mechanisms has been suggested. There are several conceivable mechanisms that lead to a resistance change of the system by purely electronic effects. One possibility is the charge-trap model,^[55] in which charges are injected by Fowler–Nordheim tunneling at high electric fields and become subsequently trapped at sites such as defects or metal nanoparticles within the insulator. This modifies the electrostatic barrier character of the MIM structure and, hence, the resistance of the structure, resembling the gate – channel resistance in Flash FET. For example, metal nanoclusters incorporated in either polymeric^[56,57] or inorganic insulator films^[58] are reported as trapping sites. Alternatively, trapping at interface states is discussed to affect the adjacent Schottky barrier at various metal/semiconducting oxide inter-

faces.^[59–61] Other possibilities of a purely electronic switching are changes in a strongly correlated electron system of transition metal oxides and/or an insulator-metal-transition (IMT), in which electronic charge injection acts like doping to induce an IMT in perovskite-type oxides such as (Pr,Ca)MnO₃^[16,62,63] and SrTiO₃:Cr.^[64] A quite generic model has been presented by Rozenberg et al.^[65,66] For all purely electronic models it has not yet been shown how the voltage-time dilemma can be overcome.^[67] Several switching models are connected to polarization reversal in ferroelectric materials. A switching model based on a tunnel junction made out of a ferroelectric material has been proposed by Esaki^[9] and theoretically described by Kohlstedt et al.^[10,68] The idea is that the tunneling current should depend on the parallel or antiparallel direction of the ferroelectric polarization. Another model is based on reasonably electron conducting ferroelectric layers in contact to conducting non-ferroelectric layers. Depending on the polarization direction, the space charge situation and the potential barrier at the interface is modified which leads to a change in the leakage through the system.^[69–72] Similar to FeRAM, the voltage-time dilemma would be overcome in these models by the voltage-induced formation of opposite ferroelectric domains and the thermodynamic stability of these domains.

In contrast to the purely electronic switching models, a large body of literature has built up during recent years in which the participation of a transport of anions is considered as essential for the resistance switching. In many transition metal oxides, oxygen ion related defects, typically oxygen vacancies, are much more mobile than the transition metal cations. An enrichment or a depletion of oxygen vacancies will affect the valence state of the transition metal cations and may lead to a considerable change in the electronic conductivity. Because of the generic character of the valence change in this class of resistive switching phenomena, we refer to this class as valence change memory (abbreviated VCM) effects. We will use SrTiO₃ as frequently studied model material in order to show the present state of knowledge of this switching mechanism, in which the redox process is induced by anion migration.

Before we start our considerations, we need to distinguish two fundamentally different geometrical localizations of the switching event, the filamentary switching scenario and the area distributed switching scenario. An example of the first scenario we encountered in the ON state of the ECM cells. Obviously, the ON resistance is independent of the electrode area in this case. In the latter scenario, the change in resistance occurs more or less homogeneously over the entire area of the electrode of the memory cell and, hence, leads to an ON resistance which is proportional to the electrode area. Using SrTiO₃ as a model material, we will first describe the fundamentals of the redox and ion migration processes, followed by examples of VCM type filamentary switching (Sec. 4.4). Thereafter, more VCM type oxide systems will be discussed including those types which exhibit area distributed switching geometries (Sec. 4.5).

4.2. Strontium Titanate as a Model System

Transition metal oxides exhibit nonstoichiometry and a related lattice disorder.^[73] The nonstoichiometry is accompanied by

mixed valencies in the cation sublattice. Because SrTiO₃ is widely studied and it shows the typical characteristics of a VCM type resistive switching oxide, we will present it as a model system to describe the lattice disorder with respect to point defects (this section) and to extended defects (Sec. 4.3). We will complement the description by referring to other oxides wherever appropriate. We will mostly refer to TiO₂ which shows a very similar electronic structure as SrTiO₃. In fact, SrTiO₃ can be regarded as two intertwined sublattices, TiO₂ and SrO, of which the TiO₂ sublattice is predominantly relevant for the electronic behavior, such as insulator-metal transitions. As we shall see, TiO₂ and SrTiO₃ belong to the class of n-type semiconducting oxides, because the Ti sublattice can easily be reduced by shallow donors (either substitutional donor-type cations, or oxygen vacancies). Acceptors always have deep energy states which can lead to only weak p-conduction. Oxides showing a strong p-conduction such as manganites will be briefly treated in Section 4.5.

In stoichiometric SrTiO₃, the conduction band is mainly formed by the empty 3d-orbitals of the Ti ions while occupied 2p-orbitals of the O ions give the primary contribution to the valence band. There are some small mutual contributions which demonstrate a weak covalency of the chemical bonds between Ti and O.^[74] The bandgap is 3.3 eV. SrTiO₃ crystallizes in a perovskite lattice ABO₃, which represents a particular favorable crystal structure, able to accommodate an extremely wide range of cations. As any other solid, SrTiO₃ must contain a finite concentration of point defects at nonzero temperatures because of the configurational entropy. Because of the densely packed crystal lattice of SrTiO₃ and other perovskite-type oxides, the most important point defects are vacancies on all three sublattices, electrons, holes, and substitutional impurities. Interstitial defects and ions on antisites play no significant role. Thus, the only feasible type of intrinsic disorder, i.e., an electrically neutral combination of lattice defects that result from spontaneous disorder, is the Schottky disorder.^[75] This conclusion is supported by theoretical calculations of defect energies.^[76] Cation vacancies are negatively charged and, hence, act as acceptor centers, while oxygen vacancies are positively charged, donor-type centers. At temperatures below approx. 1400 K, the mobility of cation vacancies in SrTiO₃ is so low^[77,78] that they can be regarded as frozen-in leading to a (small) fixed back-ground acceptor concentration. In contrast, oxygen vacancies are much more mobile and may lead to an ionic conductivity even at room temperature, in particular when extended structural defects are taken into account as fast transport paths (Sec. 4.3). Their equilibrium point defect concentration is controlled by the oxygen partial pressure *p*O₂ of the ambient atmosphere at temperatures above a threshold temperature, *T*_{exch}, according to the oxygen exchange reaction



where O₀ and V₀[•] denote oxygen ions on regular lattices sites and oxygen vacancies, respectively, according to the notation of Kröger and Vink.^[79] For details of the surface reaction kinetics and their interrelation to the bulk diffusion of V₀[•] the reader is referred to the excellent overview of Merkle and Maier.^[80] The data in this

overview allow for an estimation of the temperature *T*_{exch} below which the V₀[•] concentration can be regarded as frozen-in. For bulk samples with free surfaces, *T*_{exch} is in the order of 700 K.

Heterovalent cations can be substitutionally accommodated as dopants in SrTiO₃. If their charge is less than the host cation they substitute they act as acceptors A, if their charge is higher than the host cation they act as donors D. Ionization equilibria such as



have to be taken into account where h[•] and e' are holes and electrons. In all cases, the electronic equilibrium



has to be considered too, where ∅ represents electronic ground state of the conduction and valence bands.

Because of Reaction 10, the oxygen non-stoichiometry in SrTiO₃ can be established by annealing at approx. *T* > *T*_{exch} (so-called HT regime) under controlled oxygen partial pressure *p*O₂. The defect concentrations are calculated by solving the set of law of mass action equations of all defect equilibria. At approx. *T* < *T*_{exch} (so-called LT regime), the oxygen content can be regarded as frozen-in and the defect equilibria are determined by the law of mass action of all equilibria except Reaction 10.

By multiplying the defect concentrations with the corresponding mobilities, partial conductivities are derived. In alkaline earth titanates, the electrons and holes can be treated as polarons. The slight negative temperature dependence of the mobilities supports the large-polaron model,^[81] although the mobilities are relatively small and fall close to those expected for the small-polaron model.^[82] The electron mobility in SrTiO₃ decreases from approx. 6 cm² V⁻¹ s⁻¹ at room temperature to 0.2 cm² V⁻¹ s⁻¹ at approx. 1200 K. The hole mobility in the high temperature regime is approx. half as large. The temperature dependence of the oxygen vacancy mobility, μ_{v₀}, is determined by combining the Arrhenius law for the temperature dependence of diffusion and the Nernst–Einstein relation according to:

$$\mu_{v_0} = \frac{z_{v_0} e_0}{kT} D_0 \exp(-W_D/kT) \quad (14)$$

with the charge number *z*_{v₀} = 2 for oxygen vacancies. The activation energy of the oxygen vacancy diffusion, *W*_D, (for random walk in the bulk of the perovskite lattice) is found to be in the range of 1.0 to 1.1 eV for SrTiO₃ from conductivity studies and diffusion studies.^[83,84] For the translation of data between these two types of studies, the ambipolar nature of the chemical diffusion as well as trapping effects in the diffusion front have to be considered.^[85]

The point defect model and data discussed so far have been derived from experiments conducted at ceramic and single crystal samples prepared at very high temperatures (*T* > 1600 K). It may be speculated to what extent this model applies to thin films deposited at much lower temperatures (*T* < 1200 K) at which e. g. the cation vacancy mobility in the bulk is quite low. It might be

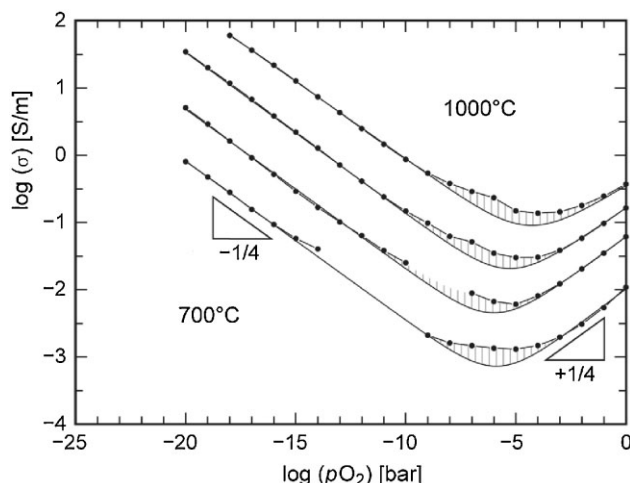


Figure 13. High-temperature electrical conductivity of an epitaxial SrTiO_3 thin film with a thickness of 1 μm grown on MgO substrates by pulsed laser deposition. The conductivity is shown after equilibration at various oxygen partial pressures $p\text{O}_2$. Reproduced with permission from [86]. Copyright 2006, Wiley-Blackwell. The conductivity is determined by electronic contributions of electrons and holes (see HT case in Fig. 14). Only the hashed areas near the conductivity minima show the deviation from the purely electronic conductivity presumably due to an ionic partial conductivity of oxygen vacancies. The temperature dependence of the electronic conductivity is given by the temperature dependence of the electron and hole concentrations, while the temperature dependence of the mobility can be neglected (see text).

possible that completely different defect scenarios are established. However, first results obtained for epitaxial SrTiO_3 thin films grown by PLD exhibit a HT conductivity behavior which is similar to that of bulk samples

The HT conductivity as a function of the equilibrium oxygen partial pressure is shown in Figure 13 of a nominally undoped epitaxial SrTiO_3 thin film grown by pulsed laser deposition.^[86] Although the growth temperatures are much lower than the crystallization temperatures during single crystal growth and sintering of bulk ceramics, the curves in Figure 13 show the same general behavior as those obtained for bulk SrTiO_3 .^[87] This indicates that the HT defect model is similar for epitaxial films and bulk crystals, in first approximation.

The curves in Figure 13 can be interpreted by a calculation which is based on the law of mass action equations of the equilibria described above. As an example, Figure 14 shows the calculated equilibrium defect concentrations as function of $p\text{O}_2$ in the HT regime (e.g., $T = 1000\text{ K}$), and, quenched at every position on the $p\text{O}_2$ axis, the corresponding defect concentrations in the LT regime (e.g., $T = 400\text{ K}$) for a slightly acceptor doped SrTiO_3 sample (possibly also a nominally undoped SrTiO_3 where the Sr vacancies act as native acceptor dopants). The typical HT conductivity minimum in Figure 13 results from the HT electron and hole concentration, n and p , in Figure 14. This includes the characteristic slopes of the HT conductivity and concentrations, respectively, of $-1/4$ and $+1/4$ at both sides of the minimum.

The different behavior in lower and higher $p\text{O}_2$ regimes result from the activation energies of the defect states. Donor states, such as the states of $\text{V}_\text{O}^\bullet$, are shallow, i.e., in the range of $<0.1\text{ eV}$ below the conduction band edge, while acceptor states lie deep in

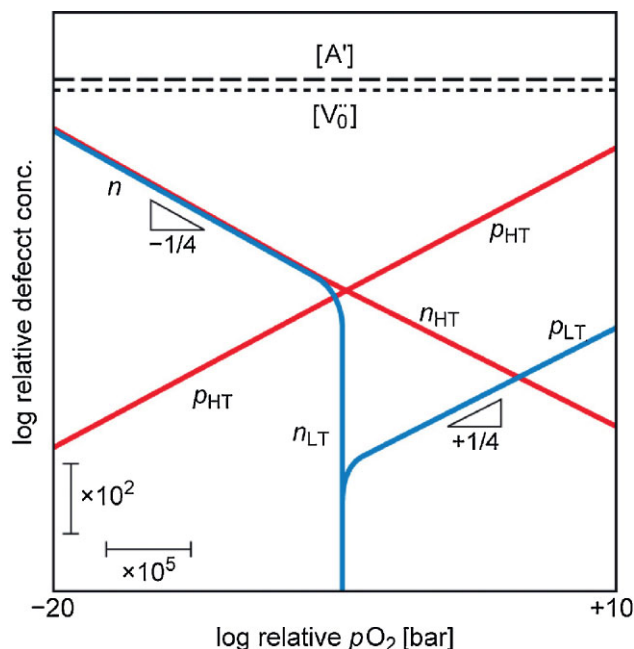


Figure 14. An idealized Kröger-Vink diagram of an acceptor-doped SrTiO_3 . The concentrations of the (singly-ionized) acceptor $[\text{A}']$, the oxygen vacancies $[\text{V}_\text{O}^\bullet]$, the electrons n , and the holes p , are shown as a function of the oxygen partial pressure $p\text{O}_2$. The concentrations of the electronic defects change drastically from the high-temperature (HT) annealing conditions to the low-temperature (LT) situation at which the oxygen content is frozen-in.

the band gap, approx. 0.5 to 1.5 eV above the valence band. As a consequence, holes are trapped during quenching from HT to LT in the oxidizing $p\text{O}_2$ regime which results in far less conductive titanates.

Interfaces such as surfaces and grain boundaries in titanates strongly affect the defect concentrations and the partial conductivities due to a high density of electronic interface states. For example, in acceptor-doped SrTiO_3 annealed under oxidizing atmosphere, positively charged grain boundary states lead to the formation of Schottky space charge depletion layers.^[88–90] At the electrode interfaces, the work function of the electrode metal strongly determines the extent of the space charge depletion layer in the titanate. For high work function metal electrodes such as Pt, pronounced depletion layers and correspondingly high impedance interfaces are formed in SrTiO_3 while low work function metal electrodes such as Ti or Al exhibit low impedance interfaces. Accordingly, the electronic charge transport through capacitor-like SrTiO_3 thin film samples with planar electrodes is controlled by a thermionic carrier emission from the cathode into the titanate where the emission current strongly depends on the work function of the cathode (Figure 15).^[91] In addition, there is a significant (sometimes dominating) impedance contribution by drift-diffusion in the bulk of the thin film.^[92–94]

In the context of the resistive switching it must be noted that the properties of titanate thin films with respect to the point defect structure established at high temperatures and with respect to the band diagram refer to *virgin* films before the electroforming has been executed. Undoped films show a high resistance at $T = 300\text{ K}$ under these conditions. Additional care must be exercised if fast pulse measurements are conducted on those

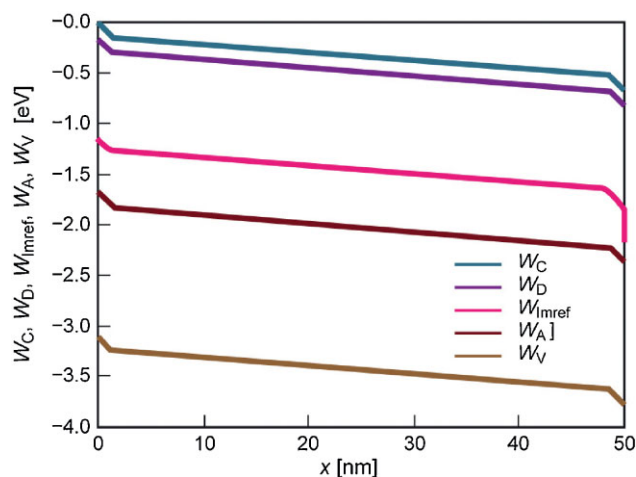


Figure 15. Band diagram of a Pt/SrTiO₃/Pt thin film system (with perfect interfaces). For the calculation of the band diagram at $T = 300$ K, a film thickness of 50 nm, an external voltage $V = 1$ V, an acceptor concentration and a compensating oxygen vacancy concentration of 10^{17} cm^{-3} , a permittivity of 300, and a Pt work function of 5.35 eV are assumed. Adapted from [93]. Based on manifold experimental findings, a low-permittivity interface layer of 2 nm thickness and a permittivity of 22 is used in the calculation.

films. As potentiostatic step experiments reveal, the dielectric displacement currents (also called dielectric relaxation currents) exceed the leakage currents for times approx. < 1 s.^[95]

4.3. Extended Defects in Transition Metal Oxides

The stoichiometry width of the SrTiO₃ phase caused by point defects is well below 0.1 at %, even at high temperatures.^[96] The same is true for many binary oxides. In TiO_{2-x}, a model material for resistively switching binary oxides, it has been found that for the accommodation of statistically distributed point defects the maximum of x is in the order of 10^{-4} .^[97,98] Above this critical value, the formation of extended defects sets in. For instance, the oxygen vacancies are accumulated into vacancy chains, and insulated crystallographic shear planes (Wadsley defects) occur.^[98] Also other types of extended defects – line defects such as edge and screw dislocations – are prone to accumulate point defects in their cores.

In fact, the ordering of point defects in a stoichiometric SrTiO₃ crystal was observed directly using ultra-high resolution TEM based on the correction of the spherical aberration.^[99] This unique technique allows for the study and analysis of oxygen vacancy concentrations along each oxygen column. For example, within a 4 nm thick specimen prepared for the TEM study (Fig. 16a) an accumulation of oxygen vacancies in two columns of approx. 15% and 30%, respectively, was observed relative to the stoichiometric value. It is astonishing that such a high degree of self-assembly of oxygen vacancies takes place extremely localized, i.e., oxygen columns with reduced oxygen stoichiometry and those with nominal oxygen stoichiometry are separated by just one unit cell. The corresponding extreme oxygen vacancy concentration gradients cannot be explained by the conventional random walk model. Ab-initio calculation of the electronic

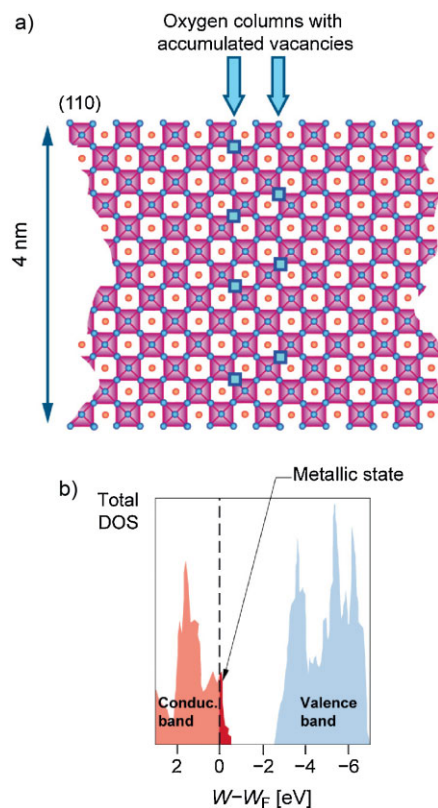


Figure 16. Accumulation of oxygen vacancies along lines in the perovskite lattice of SrTiO₃. a) Schematic cross-section of the accumulation of a thin SrTiO₃ slab prepared for HRTEM studies. Adapted from ref. [99]. Copyright 2003, Science. b) Calculated density of states (DOS) for SrTiO₃ showing the creation of Ti metallic states close to the oxygen columns with high concentration of oxygen vacancies. Calculated by Dr. G. Bihlmayer, FZ Jülich.

structure reveal that the 3d states of Ti ions in the vicinity of columns with agglomerated oxygen vacancies are delocalized and a local insulator-metal-transition can be induced (Fig. 16b).

In a real SrTiO₃ crystal these line defects would create a short circuit in an insulating matrix. As we shall see later, this kind of metallic or semiconducting “nanowires” distributed in an insulating matrix may play a very important role for the electroforming and switching phenomena. It should be noted that in contrast to dislocations, columns with reduced occupancy of oxygen can simply end within the crystal. Therefore, for an electric connection between electrodes via a fragment of conducting columns such kinds of defects should reach a percolation threshold. Diffusion coefficients for a motion along these nonstoichiometric columns in SrTiO₃ are not yet known, but intuitively they may be similar to the (large) diffusion coefficients for pipe-diffusion along dislocations.

The ordering of vacancies in SrTiO₃ also can occur, as expected, in the core of dislocations. According to the standard definition, dislocations are described by line defects which generate a disturbance in the lattice periodicity through the crystal and induce a strain field around the core. Such a change in regularity is connected with Burgers vectors. The invariance of the Burgers vector implies that the dislocation cannot simply end in the crystal. This is a very important point in the search for

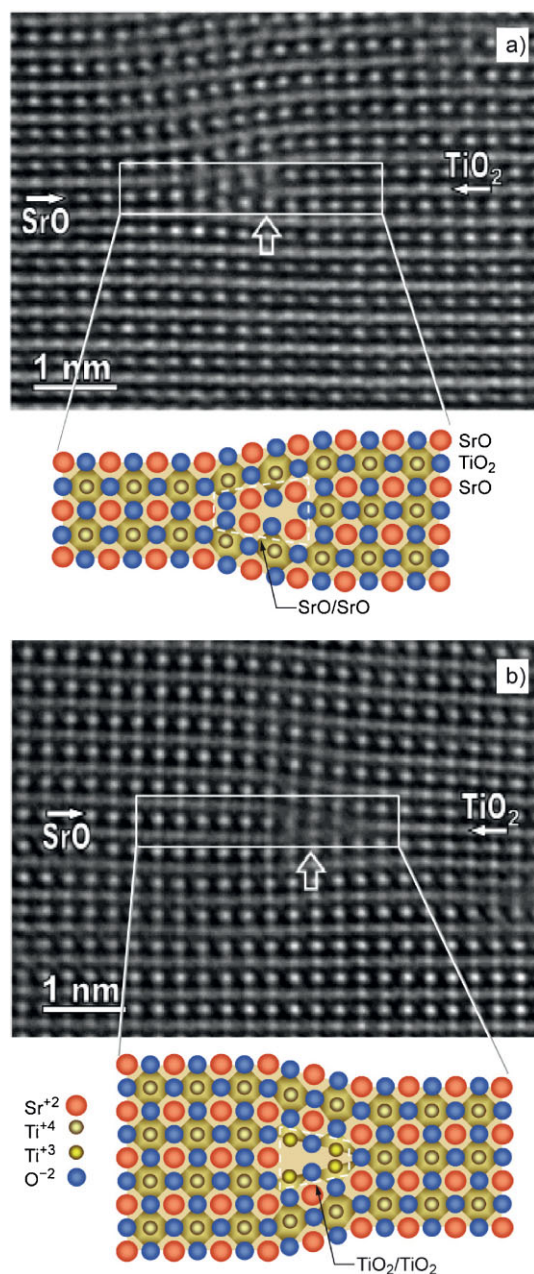


Figure 17. Schematic illustration of the crystal lattice and the site occupancy close to the core of edge dislocations in SrTiO_3 . Adapted from [101]. Copyright 2006, Taylor Francis.

defects which can connect the external electrodes parallel to the non-conducting matrix of SrTiO_3 during resistive switching events. For an understanding of the role of dislocations during the switching, we need more information about the properties of the core of dislocations. The arrangement of Sr, O, and Ti ions in the core of the edge dislocation in SrTiO_3 was studied using the HRTEM technique.^[100] The distribution of ions in the two types of cores is schematically depicted in Figure 17 which are HRTEM images obtained by Jia et al.^[102] supplemented by illustrations of the lattice.

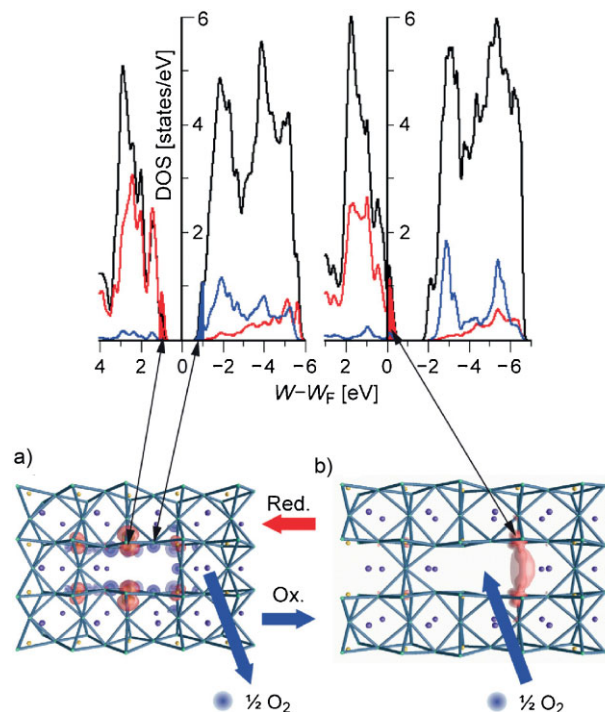


Figure 18. Electronic structure of an extended defect in SrTiO_3 as obtained by an *ab-initio* calculation. Reproduced with permission from [104]. Copyright 2007, Springer. a) Extended defect in stoichiometric SrTiO_3 . b) Extended defect in reduced SrTiO_3 . W denotes the energy and W_F the Fermi energy in the density of state (DOS) diagrams.

Dislocation cores are de facto nanostacking faults by inserted SrO and TiO_2 planes, respectively. The occupancy of the oxygen sites along the cores is reduced. As in the case discussed above, this leads to a local change of the Ti valence. This result was found not only on the basis of the analysis of the O intensity in the HRTEM image but also via a direct measurement of the Ti valence by EELS.^[100] The simulations (Fig. 18) show that close to the core of the Ti rich edge dislocation the band gap shrinks. This may lead to an increase of the conductivity of the dislocation compared to that of the surrounding matrix. Such an enhancement of the conductivity of the dislocation core in the non-conducting matrix was also observed for stoichiometric SrTiO_3 at room temperature^[104] using local conduction AFM (LC-AFM) measurements utilizing a system with an extremely high current sensitivity of about 1 fA. These results indicate that the dislocation in stoichiometric SrTiO_3 can effectively channel the flow of electronic carriers, which is important for the electroformation process.

This view of the role of dislocations is further supported by annealing studies under controlled oxygen partial pressures at elevated temperatures. It has been shown that the reduction of SrTiO_3 crystals at 950 to 1050 K leads to a fast removal (few minutes) of a tiny amount of oxygen (approx. 10^{14} oxygen ions/ cm^3). Despite this small amount, the process induces an insulator-metal transition.^[105] This concentration is approx. four orders of magnitude smaller than the concentration which is calculated on the basis of the Mott criterion for a random distribution of point defects.^[97] This extreme difference can only be understood by assuming that the influence of the reduction

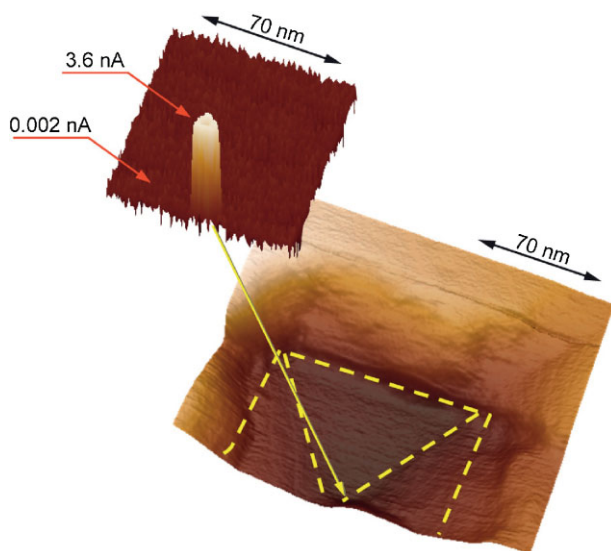


Figure 19. The position of etch pits determined by AFM topography and the coinciding position of filaments with enhanced conductivity (by LC-AFM) shown in the surface layer of the (100) face of a reduced SrTiO_3 crystal. The SrTiO_3 has been extensively etched and subsequently annealed at 1023 K for 30 min at $p\text{O}_2 = 10^{-9}$ mbar (K. Szot, unpublished results).

process is limited to the core of dislocation. Obviously, the concentration of oxygen vacancies along the dislocation cores can be controlled via an external $p\text{O}_2$, much faster than the entire bulk crystal reacts. It allows for the selective manipulation of the electronic structure along the dislocation core, i.e., a reversible change in the density of d^1 and d^2 states of Ti ions.

A specific experiment reveals additional evidence on the pronounced redox activity of dislocations in SrTiO_3 single crystals. A crystal has been etched in order to decorate dislocations by etch pits on the polished surfaces. Then, the crystal has been annealed under reducing atmosphere and, subsequently, investigated by AFM for a coincidence of the topographical etch pits and the conductivity mapping by LC-AFM. Figure 19 clearly proves a very confined and very pronounced conductivity enhancement at the center of the etch pit at which the exit of the dislocation is assumed. This indicates that the reduction process predominantly takes place along the core of the dislocation.

The fast oxygen ion transport along extended defects was further verified by a tracer experiment based on the incorporation of ^{18}O in reduced SrTiO_3 crystals at $T = 1050$ K and a subsequent determination of ^{18}O diffusion profiles by SIMS. The analysis of the penetration depth of the tracer using the Fisher–Suzuoka–Le Claire model provides evidence that the diffusion coefficient along the dislocation is about four orders of magnitude higher than that of the regular lattice.^[105] Moreover, the isotope exchange experiment indicates that the diffusion doesn't take place along insulated dislocations but along an array of dislocations. This observation has to be considered in conjunction with the TEM investigations on the change of the dislocation density in the surface region of the SrTiO_3 crystal.^[106] A decrease of the edge-dislocation density from approx. $6 \times 10^9 \text{ cm}^{-2}$ in the upper part of the surface region to approx. $2 \times 10^8 \text{ cm}^{-2}$ in the bulk of the crystal can be imagined in terms of a hierarchical array of

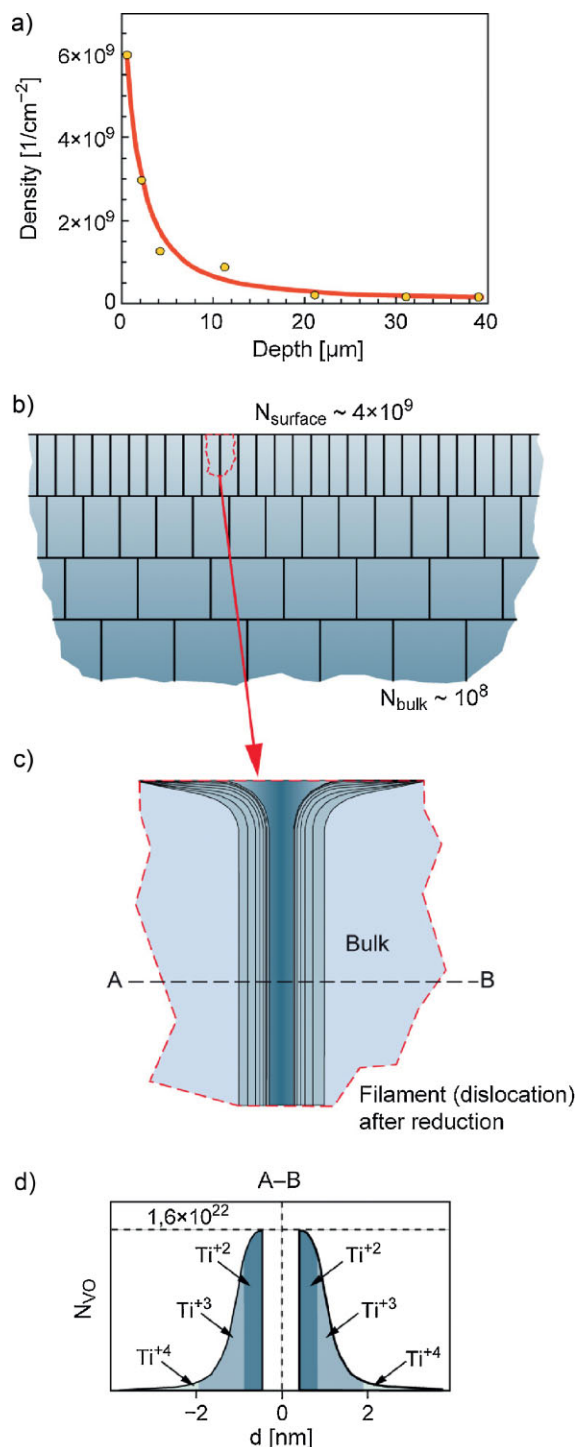


Figure 20. a) Depth dependence of dislocation density in the surface layer of an SrTiO_3 crystal determined by TEM studies. Adapted from [106]. Copyright 1998, APS. It is conceivable that the high dislocation density near the surface is due to the polishing process applied to the surface prior to the studies. b) Schematic cross-section of a hierarchical tree of dislocations in the surface layer of SrTiO_3 . c) Sketched distribution of Ti-valences in the core of dislocation after reduction (K. Szot et al., unpublished results).

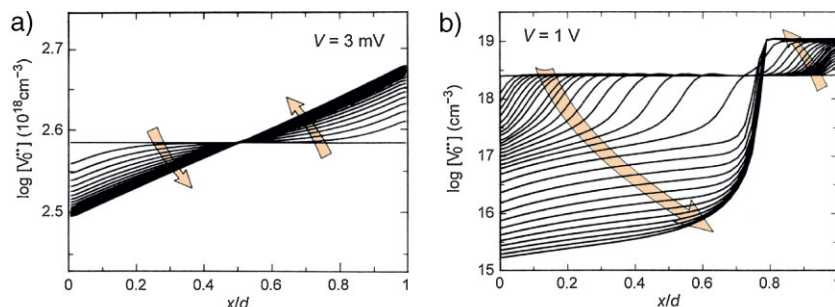


Figure 21. Calculated time evolution of the oxygen vacancy concentration profiles after applying a dc voltage to a 0.1 at % acceptor-doped SrTiO₃ crystal. a) for a low voltage, $V = 3$ mV; at steady state, $t \rightarrow \infty$, a linear gradient is obtained. For details see [108]. Reproduced with permission from [108]. Copyright 1993, Springer. b) for high voltage stress, $V = 1$ V, leading to huge gradients in the defect concentrations [109]. Reproduced with permission from [109]. Copyright 1990, Wiley-Blackwell. Please note the very different scales at the ordinate: a) a small interval on the linear scale; b) several orders of magnitude on the logarithmic scale.

dislocations (Fig. 20). It should be noted that this decrease is on a length scale which is not relevant for high-density RRAMs.

A similar distribution and density (10^{10} cm^{-2}) of conducting exits of dislocations can be observed using LC-AFM on the surface of a slightly reduced SrTiO₃ crystal. In our opinion, this preferential network of conducting dislocations plays the dominant role during resistive switching by forming a corresponding network of electronically conducting filaments. Only little is known yet about the interaction of the point defects in the matrix of the SrTiO₃ described in Section 4.2 and the dislocations (and other extended defects).

4.4. A Conceivable Forming and Switching Mechanism

In a first approach to an understanding of the **electroforming** process one may ask what happens to a mixed electronic-ionic conducting oxide, such as SrTiO₃ in the LT regime, when a dc voltage is applied to a MIM system. While the electrode interfaces M/I are considered to be sufficiently transparent for electronic carriers, the answer to this question mainly depends on the ionic interface reaction.^[107] If the electrodes are (ideally) non-blocking for the ionic partial current in the oxide, the concentration profiles between the electrodes across the oxide will not change dramatically. If, however, the electrode interfaces are blocking the ionic partial current, a typical concentration polarization occurs in the oxide. At low voltages, for which the local change in concentrations is too small to significantly affect the conductivity, a linear concentration gradient builds up (Fig. 21a) which can be described by the balance of the drift and diffusion currents according to

$$\frac{\partial [V_O^{\bullet\bullet}]}{\partial x} = \frac{\sigma_{V_O} E}{2eD_{V_O}} \quad (15)$$

where σ_{V_O} denotes the ionic conductivity due to oxygen vacancies and D is the ambipolar diffusion coefficient.

The situation changes dramatically, if larger voltages are applied and the linear transport theory cannot be applied anymore. A numerical simulation involving the continuity

equation, the Poisson equation, and the set of mass action equations leads to a significant increase of the oxygen vacancy concentration near the cathode, along with a pronounced n -conductivity in this region, and a strong depletion of the oxygen vacancy concentration in a somewhat larger anodic region, in which the p -type conduction gets enhanced compared to the initial situation.^[109] This process can be visualized by recording the optical transmission image of a slightly Fe acceptor doped SrTiO₃ single crystal under voltage stress for $T < T_{\text{exch}}^{[110]}$ (Fig. 22).^[111] The red color in the anodic region is due to a high concentration of Fe ions in the valence state +4 indicating the p -conducting region, while the white color in the cathodic region show Fe ions in a lower valence state and indicate a dominant n -conduction in this region. In fact, the pn-junction induced by the pronounced oxygen vacancy concentration polarization leads to a diode characteristic^[111] and an enhanced voltage drop in the pn-transition region.

Typically the anode does not completely block the ionic partial current, as exemplified by an oxygen release at the electrode and eventually the formation of entrapped O₂ gas bubbles.^[103,112] In this case, the n -conducting cathodic region, sometimes called “**virtual cathode**”, propagates towards the anode. This process corresponds to the electroformation. When this virtual cathode approaches and almost touches the anode, the resistance of the MIM system decreases significantly (usually limited by a compliance current) and the electroformation is completed.

The picture presented so far is simplified because it does not take into account the structure of the SrTiO₃ crystals on a mesoscopic scale. A magnification of the electrocoloration front

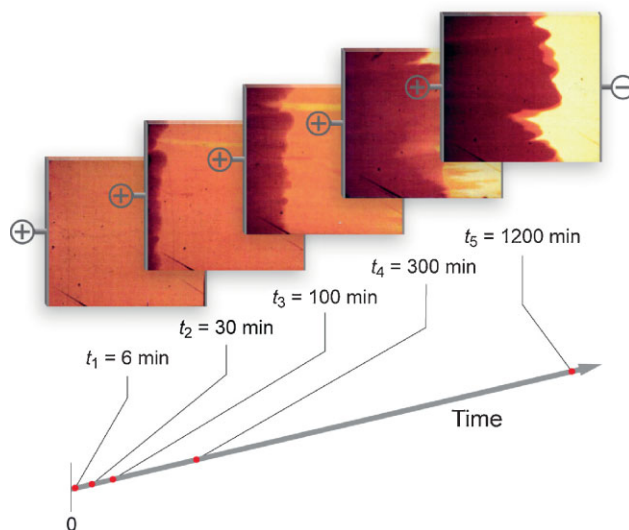


Figure 22. Optical micrographs in transmission light showing the time evolution of the electrocoloration of a 0.15 at % Fe-doped SrTiO₃ crystal (5×5 mm, thickness 0.5 mm) at $T = 453$ K and $E = 1 \text{ kV cm}^{-1}$. The crystal has been initially annealed in oxygen at 973 K for 1 h. Electrodes at the edges of the crystal were prepared by burning in Pt paste [110].

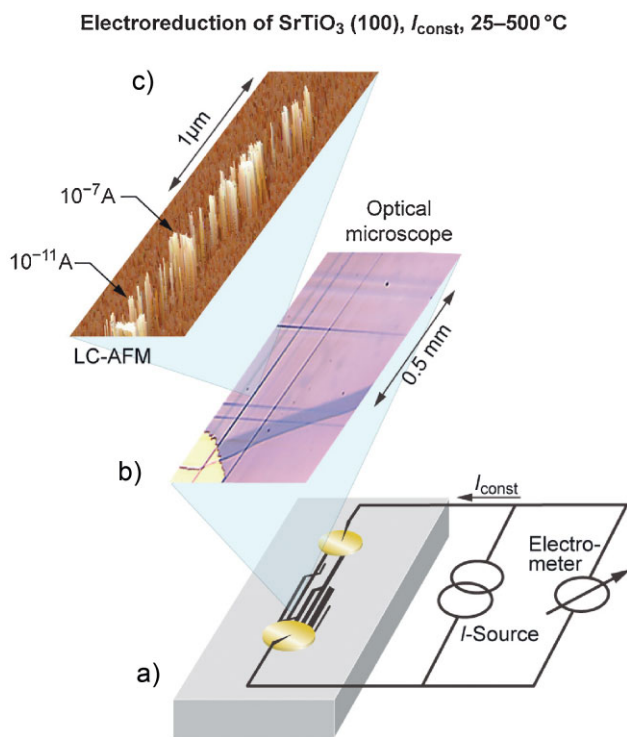


Figure 23. Electroformation of SrTiO_3 single crystals showing an orthogonal structure of filaments induced in the surface layer of SrTiO_3 (100). a) Sketch of the electrical circuit for the measurement of resistivity of SrTiO_3 crystal in the time domain at $I = \text{const}$. b) Optical image of the filaments network close to the cathode. c) Conductivity distribution measurement (LC-AFM) along the filaments which cross the surface of the SrTiO_3 crystal. Redrawn from [104].

shows its inhomogeneity. The coloration proceeds in stripes which follow simple crystallographic orientations. A closer examination shows that these stripes consist of bundles of filaments which are obviously formed by extended defects.^[103] Figure 23 shows an optical micrograph and a LC-AFM image of a SrTiO_3 during the electroformation process. These findings clearly indicate the relevance of extended defects as fast ion transport paths in the LT regime.

A spectroscopic study complements the findings described above. Janousch et al.^[113] have investigated the electroforming procedure of lateral cells fabricated on 0.2 at % Cr-doped SrTiO_3 single crystals by infrared thermal microscopy, micro X-ray fluorescence (XRF) and X-ray absorption near-edge spectroscopy (XANES), set into the context of related studies by an excellent review paper of Karg et al.^[114] Figure 24a shows the XRF map taken at the Cr pre-edge region of a Cr-doped SrTiO_3 single crystal after electroforming. By comparing the data with thermally reduced single crystals, it was concluded that the pronounced Cr pre-edge intensity can be attributed to the existence of oxygen vacancies. Therefore, the color scale of Figure 24a represents the amount of oxygen vacancies in the Cr octahedra. In first approximation, the result is consistent with the electromigration process described above. It can be clearly seen in the figure that the forming process introduced a few micrometers wide region of high oxygen vacancy concentration across the single crystal between the cathode and the anode. Additional information is

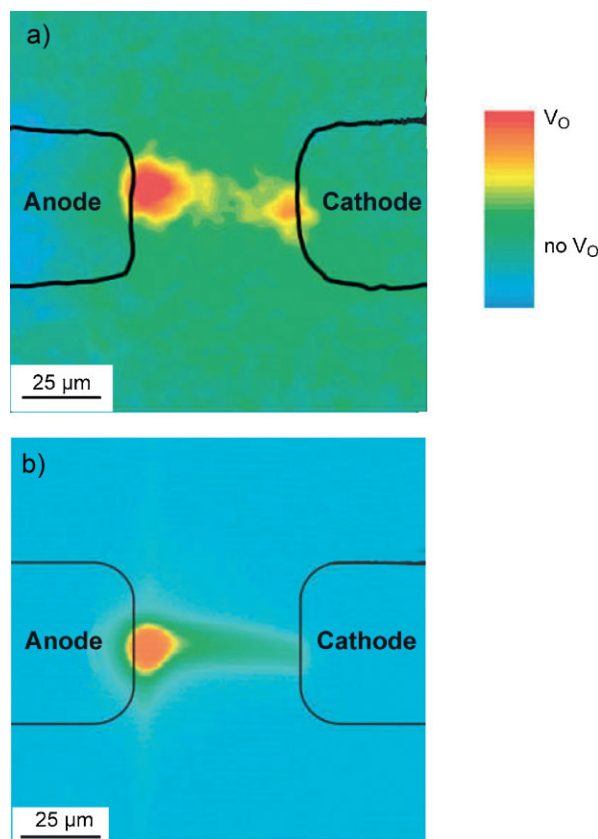


Figure 24. Electroformation of a lateral cell on top of a Cr-doped SrTiO_3 single crystal. Reproduced with permission from [113]. Copyright 2007, Wiley-VCH. a) Cr XRF map taken at 6004.3 eV for maximum contrast at the Cr pre-edge region. In the color scale, red represents oxygen vacancies in the Cr octahedral position of the perovskite cell. b) Infrared thermal image of the cell with a total power dissipation of approx. 150 mW. In the color scale, blue and red represent room temperature and elevated temperature, respectively.

provided by an infrared image recorded with an applied voltage of 30 V (of the same polarity as used for the electroforming process) as shown in Figure 24b. The power dissipation is clearly located at the anodic interface while there is no specific temperature increase at the cathode. This is consistent with our description of a virtual cathode made from an oxygen vacancy enriched region propagating towards the anode where the electrical field becomes concentrated. As will be described in more detail below, the anode during electroforming typically acts as the active interface during the subsequent resistive switching process.

Jameson et al.^[115] confirmed this model for TiO_2 single crystals and complemented it by electrical measurements. They have investigated the influence of an applied forming voltage on the Schottky barriers formed at the two interfaces of lateral Pt electrodes on the surface of rutile single crystals. A unique dependence on the crystallographic orientation of the TiO_2 single crystal demonstrated preferred, crystallographically aligned transport channels for the oxygen vacancies ruling out a migration mechanism based on a random walk of oxygen vacancies.

The electroforming conditions strongly depend on the dimension of the sample, in particular, the electrode thickness. While single crystals typically require some 10 V to 100 V for

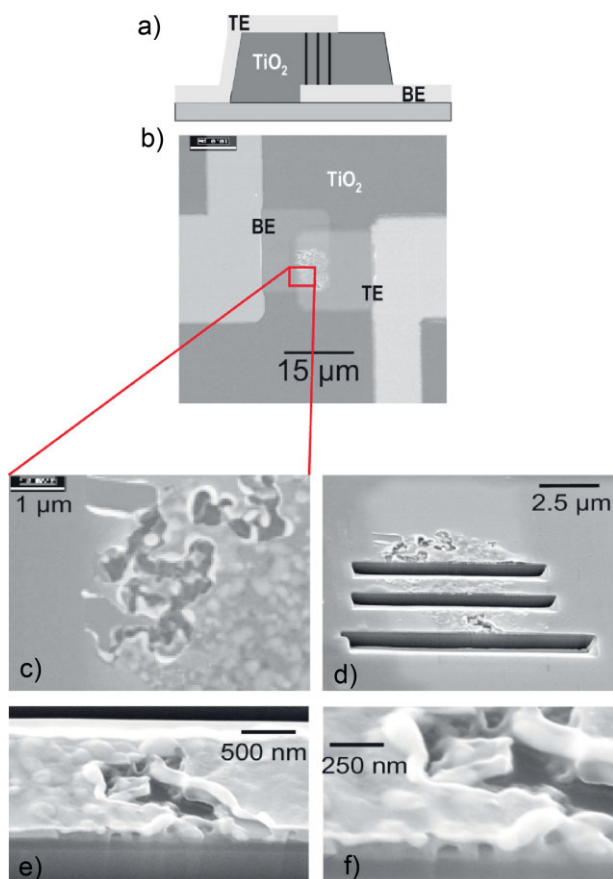


Figure 25. SEM images showing the morphological change as a results of the electroforming process of a Pt(10 nm)/TiO₂(27 nm)/Pt(10 nm) cell. Reproduced with permission from [119]. Copyright 2007, Elsevier. a) sketch of the cell; b) overview of the cell; c) morphologically modified area; (d) focussed ion beam (FIB) cuts; e) and f) magnified cross-sections of the FIB cuts.

many minutes or hours, for thin film samples, a first cycle of few seconds and an amplitude of one or few volts is often sufficient.^[116,117] In the case of thin electrocatalytical electrodes such as Pt exposed to the ambient atmosphere, the oxygen partial pressure also plays a role.^[118] In addition, thermal effects seem to play an essential role in the electroforming. Whenever forming (or switching) currents are in the range of some tenths of a mA or higher, morphological changes may occur if the current load is confined to a small area of a thin film, e.g., at an extended defect which acts as a dominant paths of the current, presumably because of thermal damage.^[119] Figure 25 shows the morphological changes which have been detected for a sputtered TiO₂ film of 27 nm thickness with a 10 nm Pt top electrode subjected to a forming process with a compliance current of 0.1 mA. The thin Pt top electrode is coagulated over an area with a diameter of few μm. Focused ion beam cuts prove that the morphological changes are not restricted to the Pt electrode but included the formation of nanometer sized pores in the TiO₂ film too. Details can be found in ^[118,120]. Still it is not clear to what extent these morphological changes are interlinked with a particular type of forming and switching events. It is conceivable that these changes

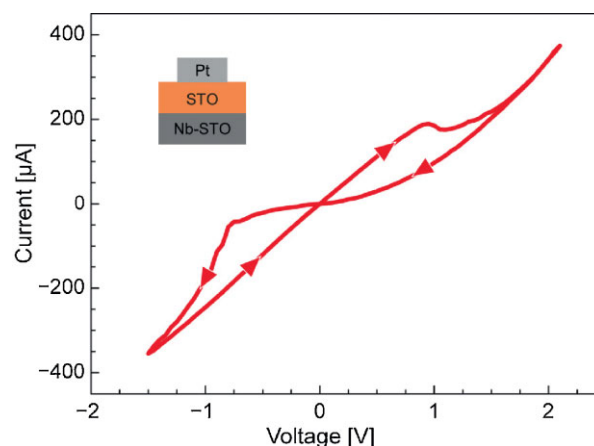


Figure 26. Bipolar switching of an epi-SrTiO₃ thin film (thickness of 500 nm) grown on a Nb-doped SrTiO₃ substrate as the bottom electrode and a Pt top electrode (diameter of 200 μm). Prior to the switching the system has been formed at +7 V for 160 s. Reproduced with permission from [121]. Copyright 2009, AIP.

are related to changes observed in cells based on the thermochemical mechanism (cf., Sec. 5)

The completion of the formation process is in any case typically indicated by a sudden increase in the current. In the framework of the electrocoloration process, this occurs when the virtual cathode reaches the anode. Despite this current increase, i.e., resistance decrease, the cell may still be in the OFF state with respect to the subsequent bipolar switching. Therefore, three resistance levels have to be distinguished: (1) virgin, before forming, (2) OFF-state, (3) ON-state. After a cell has been formed, switching between the ON and OFF states can be accomplished by applying voltage signals of opposite polarity and sufficiently high magnitude to the cell. There are many indications for a scenario in which the anode during the forming procedure represents the active electrode, i.e., the contact at which the switching takes place. Typically, the cell switches into an ON state by applying a negative voltage to the active electrode, while a positive voltage signal results in the OFF state. Figure 26 shows the switching curve of an epitaxially grown SrTiO₃ thin film with a Pt electrode as the active electrode. It should be mentioned that also a reverse switching polarity can be established in many cases which results in a stable switching behavior.

As a summary of many reports about bipolar switching in oxides, one can distinguish between an active electrode interface at which there is the dominant potential drop (in the OFF state) and an inactive, ohmic or quasi-ohmic electrode interface. This electrode asymmetry is set up either by the choice of the materials, by the geometry, or by the pre-processing such as the forming process. Some typical cases are collected in Figure 27.

In the case of LC-AFM experiments, the quasi-point contact of the tip establishes the active electrode while the counter electrode which is usually a large bottom electrode shows a quasi-ohmic behavior. Based on this view, typical LC-AFM results can be interpreted.^[103,123]

The electroforming process, including the oxygen evolution at the anode, gives rise to a virtual cathode made from oxygen vacancy rich regions emerging at the real cathode, as discussed above. This process introduces the required asymmetry into

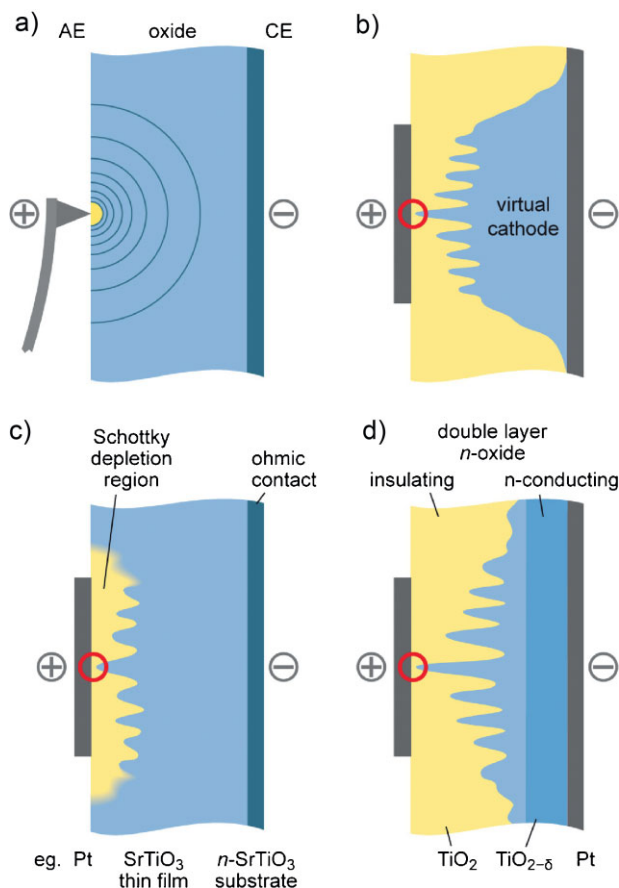


Figure 27. Sketched cross-sections of various configurations for bipolar switching oxide cells indicating how the actively switching electrode interface (AE) is formed to determine the polarity of the switching. An *n*-type oxide such as TiO_2 or SrTiO_3 and a filamentary switching are assumed in these cases. The situation after electroforming and the cell in the OFF state are shown. a) AFM tip as active electrode. The oxide, e.g., undoped SrTiO_3 single crystal, first is thermally reduced to obtain an *n*-conduction. Reproduced with permission from [103]. Copyright 2006, Nature Publishing group. A positive tip polarity induces the OFF state. b) A virtual cathode has been grown from the cathode through an extensive electroforming treatment. This turns the left electrode into the active electrode (AE). c) the polarity of the cell is determined by the electrode materials. A high work function electrode (e.g., Pt) generates a Schottky-type depletion region through which conducting filaments can grow during electroforming. Reproduced with permission from [121]. Copyright 2009, AIP. The counter electrode is made from an ohmic contact such as *n*-doped SrTiO_3 single crystal substrates or a low-workfunction metal such as Ti. d) Symmetric electrodes such as Pt can be used if a compositional gradient is built into the oxide layer. For example, an O-deficient *n*-conducting TiO_2 layer turns the right electrode into an ohmic counter electrode, while the stoichiometric layer forms the active electrode interface with the left Pt electrode. Reproduced with permission from [122]. Copyright 2008, IOP. In all configurations (b) to (d) the frontmost filament of the virtual cathode will perform the switching action (red circle).

initially symmetric MIM systems such as $\text{Pt}/\text{TiO}_2/\text{Pt}$ cells. In many cases, this process advances along extended defects in the lattice as fast migration paths. The anodic interface during electroforming establishes the active electrode for the subsequent switching process.

Electrode materials can be selected appropriately in order to establish an ohmic contact (by a low work function metal such as

Ti, or by using *n*-conducting STO:Nb substrates) and a blocking contact (by a high work function metal such as Pt or Au) at *n*-type semiconducting transition metal oxides (i.e., oxide which are easily reduced such as TiO_2 and STO). The active interface is build by the blocking contact.

A gradient can be built into the 'I' layer of the MIM cell, in order to make one electrode more blocking than the other. An impressive example has been provided by [122], in which an O-deficient $\text{TiO}_{2-\delta}$ layer has been reported to build a quasi-ohmic contact with Pt electrodes while a second, stoichiometric TiO_2 layer forms the blocking contact to the opposite Pt electrode.

The active electrode represents an electrostatic barrier, e.g., a Schottky barrier, which is modified by the applied voltage. Once the active electrode is identified, an important question concerns the polarity of the bipolar resistive switching. Most papers report a SET process upon a negative polarization of the active electrode contact and a RESET process upon positive polarization. This observation can be explained by an attraction of (positively charged) oxygen vacancies which act as donors and, therefore, reduce an electrostatic barrier in an *n*-type semiconductor. We explained this for our tip-based switching of SrTiO_3 single crystals^[103] (Fig. 28), and it has been confirmed for various MIM type cells.^[19,122]

A comment is required at this point with respect to the seemingly antithetic role of oxygen vacancies during electroforming and switching. During electroforming, an electrochemical oxidation at the anode may occur, which leads to the evolution of molecular oxygen according to Reaction 10 and the injection of oxygen vacancies into the oxide, which are immediately drawn towards the cathode by the electrical field. In contrast to some notes in the literature, oxygen vacancies do *not* accumulate under the anode, but under the cathode. Because oxygen vacancies are donors, the region in which they accumulate becomes (highly) *n*-conducting. Because of this increased conductivity, the electric field decreases significantly in this region, leading to a decrease in the oxygen vacancy motion. As a result and as mentioned above, the oxygen vacancies accumulate in the virtual cathode region which propagates from the real cathode towards the anode. As we have seen, this process mainly proceeds along extended defects. Once the fore-most virtual cathode filament comes into close vicinity of the anode (presumably few nm), the current rises significantly, possibly due to the onset of tunneling. Still, there is an electrostatic barrier between the (high work-function) metal and the tip of the virtual cathode. Figure 28 illustrates the oxygen vacancy concentration profile $[\text{V}_\text{O}^\bullet](x)$ along the switching filament assumed in our model as well as the band diagram $W(x)$. For the OFF state, the electrostatic barrier is given because the virtual cathode (its filamentary tip, along an extended defect such as a dislocation) cannot contact the active electrode since the (positively charged) oxygen vacancies are repelled from the immediate vicinity of the *positively* polarized metal. If a *negative* voltage pulse is applied to this active electrode, some oxygen vacancies are now attracted from the "tip of the virtual electrode" to the metal contact significantly narrowing the electrostatic barrier and turning the contact into the ON state. The electron transmission will be high because of a narrow barrier and a Schottky lowering of the barrier height, which leads to a quasi-ohmic *I*-*V* characteristic. Obviously, this attracting and repelling of oxygen vacancies between the metal contact and the

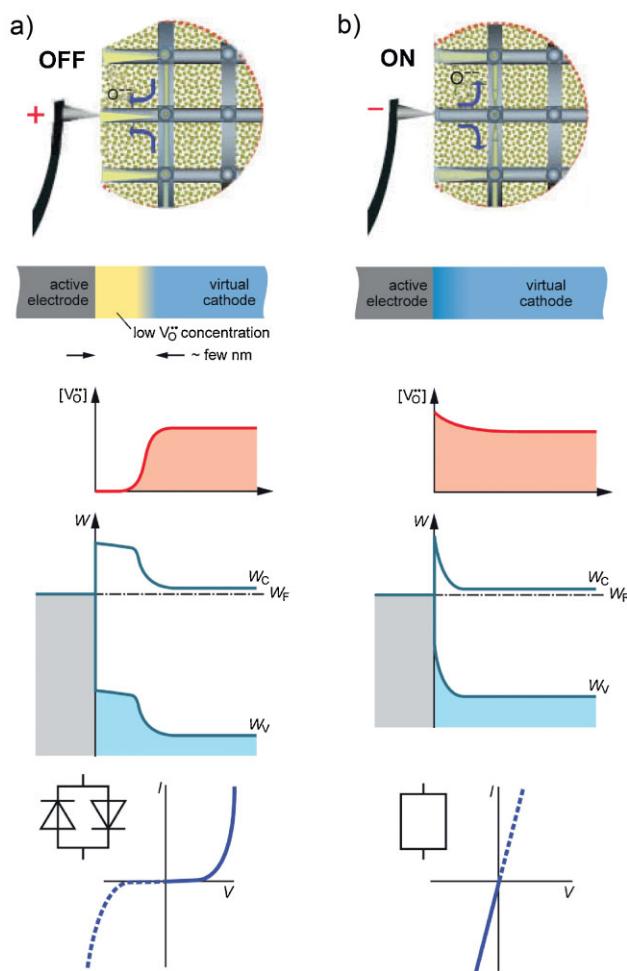


Figure 28. Sketch of the bipolar resistive switching by a LC-AFM tip at the surface of a formed SrTiO_3 single crystal, reproduced with permission from [103]. Copyright 2009, AIP. The bluegrey network illustrates the extended defects which show n -conductivity due to accumulated oxygen vacancies. a) RESET switching into the OFF state by positive polarity repelling oxygen vacancies and, thus increasing the electrostatic barrier again (with a blocking, e.g., diode characteristic). b) SET switching into the ON state by negative polarity attracting oxygen vacancies which turn the contact into an ohmic characteristic. The concentration profile of oxygen vacancies $V_O^{\bullet\bullet}$ along the filament under contact is sketched below. In addition, an illustration of the band diagram is given. W , W_F , W_C , and W_V denote the energy, the Fermi energy, the energy of the conduction band edge and of the valence band edge, respectively. Please note that the band diagram can only provide a coarse-grained picture. Because of the nanometer size diameter of the extended defects ab-initio calculations (such as in [103]) are required to provide more precise information.

front of the virtual cathode over a short distance by negative and positive voltage pulses, respectively, constitutes the core process of the bipolar switching. Strictly speaking, the band model of semiconductor physics cannot be applied anymore for low dimensional, nanosized structures such as dislocations and extended defects. However, it may still serve as a coarse-grained picture of the energy profiles. A more precise picture is provided by ab-initio methods as shown, for example, in Figure 18.

Finite-element simulations have been performed by Strukov et al.,^[124] by Nian et al.,^[125] and by Jeong^[120] to describe

characteristic signatures of the resistive switching using an 1-dimensional model of a MIM cell with appropriate boundary conditions. The simulation is based on numerical solutions of coupled drift-diffusion equations for electrons and oxygen vacancies, similar to the approach reported above.^[109]

The picture drawn in this Section is simplified, of course, and incomplete. There are several experimental results which either do not fit into this working hypothesis because they show contradicting signatures or, at least, require additional explanations. For example, a prolonged formation process may lead to an ON state of the cell, instead of an OFF state as described above. In addition, the role of the electrodes, namely the active electrode and the counter electrode, are often not defined if one of the four cases in Figure 27 is not clearly established. For symmetric MIM cells, the switching may take place alternately on both electrodes or simultaneously on both electrodes with different magnitudes.^[66,103,126]

4.5. Other Systems and Localization of the Switching Phenomenon

At this point, we return to categorizing the geometrical localization of the switching event. Obviously, this issue is not only highly relevant for the understanding of the microscopic switching process but also for the scaling potential of future non-volatile memory devices utilizing the VCM type switching.

As discussed above, in many cases there are clear indications for a switching process based on a single filament. The most obvious proof of a filamentary electroforming and switching is given by LC-AFM scan which have shown the very localized conduction at the surface of, e.g., SrTiO_3 single crystals (see above and [103]) and thin films^[123] as well as for TiO_2 thin films prepared by atomic layer deposition ALD.^[127] The SET and RESET switching of an individual filament at the surface of a SrTiO_3 single crystal is shown in Figure 29. During LC-AFM scan over the area in which an exit of a dislocation forms a conducting filament, the resistance of the filament can be switched between an ON and OFF state by a voltage applied to the tip of the LC-AFM. As has been shown in [103], the diameter of the switching filament can be as small as approx. 1 nm.

In addition, in MIM cells with single switching filaments usually the ON resistance (and often the OFF resistance too) is independent of the electrode area. If the active electrode is cut into several parts after forming and switching, only one of the parts shows the switching phenomenon while the other parts behave approximately virgin.^[122] This clearly demonstrates that after electroforming typically *one* single filament exists that is switched ON and OFF by the bipolar pulses. If this (tiny) area on the electrode is isolated, the rest of the films shows virgin I-V behavior. For large electrodes, the $R_{\text{OFF}}/R_{\text{ON}}$ ratio for VCM system often has been relatively small (e.g., [116]). This seems to be due to the fact that filamentary based switching only affects an extremely small portion of the entire electrode area and the remaining electrode area contributes to a (non-switching) parallel resistance. As a consequence, the $R_{\text{OFF}}/R_{\text{ON}}$ ratio significantly improves when the electrode area is reduced, on the condition that R_{OFF} is proportional to the area. In this respect, filamentary VCM cells show excellent scaling prospects. Quantitative data

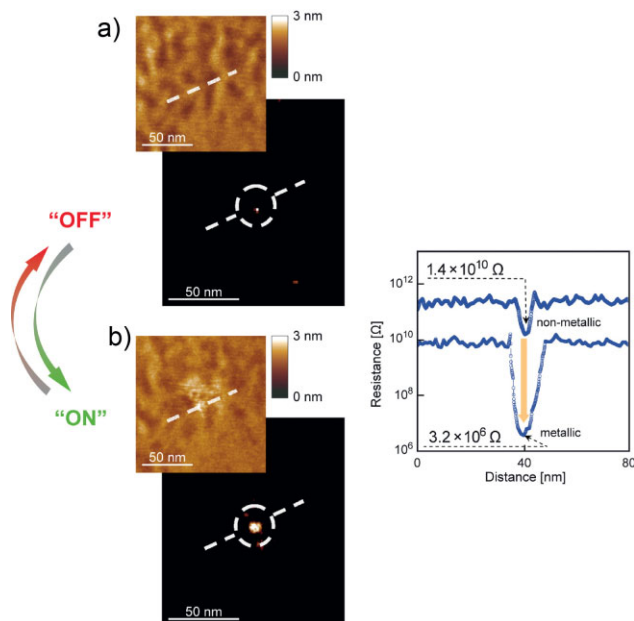


Figure 29. A conductivity map of the surface of a SrTiO₃ single crystal in the vicinity of a conducting filament as recorded by LC-AFM. The arrows (left) indicate that the filament can be switched between an ON and OFF state by applying appropriate bias voltages to the LC-AFM tip. The resistance profiles are line scans before and after the first SET scan [104]. The RESET of a single filament has been shown in [123]. Reproduced with permission from [104] and [123]. Copyright 2007, Springer, and 2007, Wiley-VCH, respectively.

based on simple assumptions have been shown by Kinoshita et al.^[128]

Besides the reports about filamentary-type resistive switching, however, there are some reports in the literature about resistive switching material combinations which show electrode area dependent resistances. Sawa^[19] reported on a so called homogeneous interface-type switching, which is observed at the interface between different conducting oxides and metals. Generally, this type of switching cells shows a clear scaling of the device resistance with the area. For this type of devices, the change of the resistance is attributed to the field-induced change of the Schottky-barrier at the interface homogeneously over the entire electrode area. Figure 30 (black curves) shows R_{ON} and R_{OFF} for cells based on Nb-doped SrTiO₃ single crystals which have been etched by a low pH solution prior to the application of a high work function top electrode metal such as Pt to form a Schottky barrier. Clearly, both resistance states exhibit values which are approximately proportional to the top electrode area over several orders of magnitude.

In addition to the two categories, the single filament type switching and the homogeneous interface type switching, which represent the extreme cases, there are several types in between. One type is the spotty type, which has been observed by LC-AFM at epitaxial SrTiO₃ thin films grown by pulsed laser deposition at relatively high growth rates.^[130] Patterns of relatively large regular spots of approx. 30 nm diameter at a relatively uniform distance have been observed, which seem to be related to extended defect configurations generated by the specific growth conditions. In addition, there are frequent reports in the literature which claim a

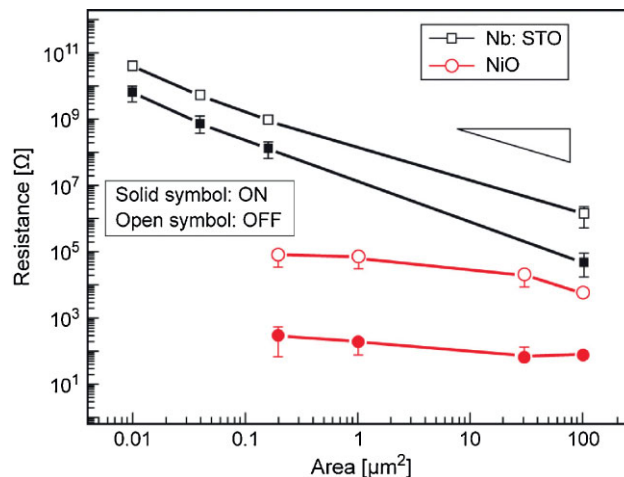


Figure 30. Area dependence of resistance values in the ON and OFF states for Nb-doped SrTiO₃ (Nb:STO) and NiO memory cells. The resistance of Nb:STO memory cells depends linearly on the area, suggesting that the resistive switching takes place over the entire area of the interface. The resistance of the TCM-type NiO memory cells is almost independent of the area, suggesting that resistive switching is a local, filamentary phenomenon. Reprinted with permission from [129]. Copyright 2005, IEEE.

multifilament type switching. Of course, in MIM structures the real category can not be distinguished between a homogeneous interface type, a spotty type, and a multifilament type unless the electrode size is varied and reduced to a size below the typical size of the spots in the spotty type or the medium distance between filaments in a multifilament type.

At this point it is useful to give an intermediate summary. In Section 4.4 we have sketched a model of the VCM type resistive switching mechanism of *n*-type oxides such as SrTiO₃ and TiO₂ as examples. The model is based on the electromigration (drift-diffusion) of oxygen vacancies $V_{\text{O}}^{\bullet\bullet}$ and a reduction of the valence of transition metal cations (here: Ti^{+z}) in regions of an $V_{\text{O}}^{\bullet\bullet}$ enrichment which leads to an enhanced *n*-type conduction. The voltage driven accumulation or depletion of $V_{\text{O}}^{\bullet\bullet}$ within the oxide in front of the active electrode leads to a modification of the Schottky type energy barrier and a corresponding switching into the ON and OFF state, respectively. This picture is further supported by results for various active electrode materials with different work functions such as Pt, Au, SrRuO₃ (SRO), and Ti.^[19,60,121,131] The contact resistance M/SrTiO_3 increases with increasing work function of the active electrode metal *M*. While a cell with a low work function metal such as Ti shows a low resistance, an electrode made from Au, SRO, or Pt shows a rectifying *I*-*V* characteristics and resistive switching. An example is shown in Figure 31a. It must be mentioned that besides the work function also the oxygen affinity of the active electrode material plays a crucial role. For example Ti shows a very high oxygen affinity which will lead to an additional depletion of oxygen ions in the adjacent oxide near the interface and, hence, an additional decrease of the barrier supporting the low contact resistance of the Ti/SrTiO₃ junction. A modification of the doping level on the SrTiO₃ side of the interface acts accordingly.^[132] For example, a layer of few unit cells of a highly *n*-conducting (La_{0.25}Sr_{0.75})TiO₃ strongly reduces the barrier and leads to a low contact resistance.

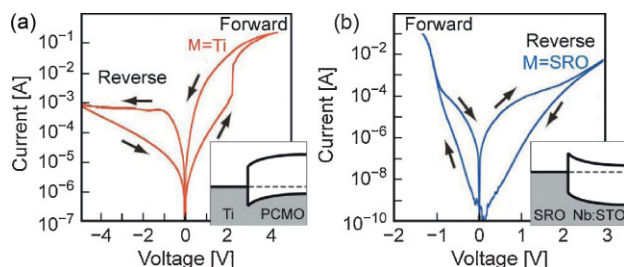


Figure 31. I - V characteristics of the contact between an active electrode M and a transition metal oxide system showing bipolar switching [19]. The top electrode is at the reference potential in these experiments. a) n -type oxide Nb-doped SrTiO₃ with M = SrRuO₃; b) p -type oxide (Pr,Ca)MnO₃ with M = Ti. The insets show the energy band bending across the contact. Copyright 2008, Elsevier.

Interface-type switching is also observed in cells made from p -type oxides such as Pr_{0.7}Ca_{0.3}MnO₃ (PCMO). In p -type conducting PCMO cells, the contact resistance between PCMO and M increases as the work function of M decreases. In opposition to the results found for n -type cells, the resistance decreases with increasing oxygen vacancy concentrations and the energy barrier for hole transport (inset in Fig. 31b) increases. As a result, also the polarity of the resistive switching is reversed compared to the n -type oxide cells. Consistent with this picture, resistive switching at the interface of Pr_{0.7}Ca_{0.3}MnO₃ is only observed for samples with a low work function Ti top electrode and not for SrRuO₃, Pt, Au, Ag^[133] (Fig. 31b). According to their

work functions, one would expect a similar behavior for Ag and Ti. The discrepancy is attributed to the high oxygen affinity of Ti, i.e., its tendency to get the oxygen from the PCMO surface. As a result, a more pronounced Schottky barrier is formed between the oxygen deficient PCMO layer and the Ti electrode. It has been shown that the hysteresis becomes less pronounced when the samples are annealed in air and the oxygen vacancy concentrations at the interface are reduced. However, the exact mechanisms of the field-induced change of the Schottky-barrier is still not clarified unambiguously, since a pure oxygen ion diffusion mechanism is often in contradiction to the observed switching polarity in interface-type switching cells.

The approach of a varying Schottky barrier height by the choice of the oxide/metal combination and the accumulation and depletion of oxygen vacancies can be extended by introducing a defined barrier layer at the interface, which enables the control of the device properties by the interlayer thickness.^[19] Sawa et al. introduced an ultrathin epitaxial Sm_{0.7}Ca_{0.3}MnO₃ layer at the Ti/La_{0.7}Sr_{0.3}MnO₃ interface and were able to control the R_{ON}/R_{OFF} ratio by its thickness.^[59] A similar concept utilizes a thin layer of a transition metal such as Sm which becomes oxidized and (partially) reduced again by a positive and negative voltage applied to the adjacent electrode in a Mo/SmO_x/(La,Ca)MnO₃/Pt cell.^[134]

Another successful approach towards a design of resistive switching cells with tunable properties was presented by Meyer et al.^[135] They introduced a thin oxide tunnel barrier (TO) at the interface between Pt and a conducting metal oxide (CMO). The memory cell is based on the field induced oxygen ion transfer

between the conductive oxide, which acts as a mixed ionic electronic conductor, and the insulating tunneling oxide whose effective barrier height can be modulated by the oxygen ion content. Resistance changes in the memory cell can be understood in terms of a variation of the space charge in the tunnel barrier which is provided by oxygen ions (Fig. 32a), resulting in a corresponding variation of the height of the tunneling barrier. A positive voltage of 2–3 V results in the OFF state of the cell. Oxygen ions are attracted from the adjacent region of the CMO into the TO resulting in an increase of the effective tunnel barrier height and a corresponding increase of the resistance (OFF state). The depth from which the oxygen ions are attracted in the CMO is given by the Debye length which describes the penetration depth of the electric field into the CMO. Because a CMO with a high conductivity is selected for this concept, the Debye length is very short (few nm). This is important for the kinetics of the process (see Sec. 4.6). A negative voltage of 2 to 3 V leads to a SET process and an ON state of the cell. The polarity observed for the SET and RESET process of this type of cell is obviously equal to the polarity of the n -type oxide cells described above.

Excellent scaling of the read and write currents proportional to the electrode area and

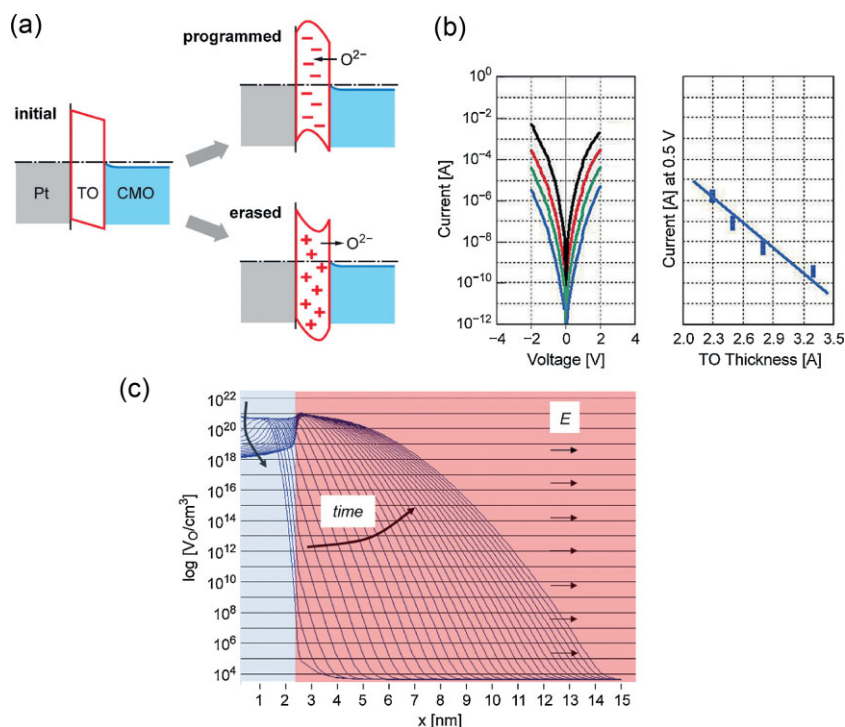


Figure 32. Active electrode interface of the oxide dual-layer memory concept. a) Sketch of the potential barrier for initial, programmed (RESET) and erased (SET) memory cell. b) Cell characteristic as a function of tunnel oxide thickness and scaling of the current with tunnel oxide thickness. c) Simulated evolution of the oxygen vacancy profile with time during "programming" (RESET). Courtesy of Unity Semiconductor Corp. Reproduced with permission from [135]. Copyright 2008, IEEE.

an exponential dependence of the currents on the barrier thickness (Fig. 32b) indicate a homogeneous interface-type switching rather than a filamentary switching mechanism. Numerical methods are employed to understand the oxygen motion across the interface in more detail. Figure 32c shows the evolution of the oxygen vacancies profile during the write operation calculated by solving the non-linear transport equations and the continuity equation for oxygen vacancies, oxygen interstitials and electronic charge carriers.

Even though the interface-type resistive switching cells described above show improved device homogeneity compared to resistive switching devices with pronounced filamentary character, it cannot be concluded that the movement of ions and electrons occurs completely homogeneously over the whole device area. In contrast to the assumption of a homogeneous charge transport, it has been demonstrated for PCMO^[136] as well for Nb-doped STO^[130] by LC-AFM that the conductivity of these materials is strongly inhomogeneous. Due to the statistical distribution of these inhomogeneities on the nanoscale, the properties of the above mentioned devices may scale with the area for device dimensions in the μm range, but may differ from the scaling behavior in the nanometer range. Thus, extended device scaling experiments down to the nanometer scale have to be performed for clarification.

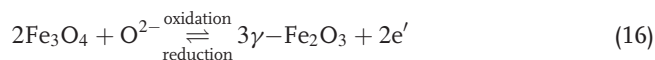
4.6. Switching Kinetics and the Voltage-Time Dilemma

Several papers report switching times below 10 ns. This is confirmed by our studies using Pt/TiO₂/Pt cross-point cells with a cross-section of 100 nm $\times$ 100 nm.^[137,138] An estimation based on the high-temperature oxygen vacancy diffusion data extrapolated to 300 K and application of the Nernst–Einstein relationship shows that oxygen vacancies can hardly migrate even 1 nm during the switching time under conventional (random walk) conditions. There must be one or more accelerating factors. One possibility in the case of the filamentary mechanism is the faster transport rate along extended defects. As described in Section 4.3, there are many indications pointing in this direction. Another possibility is thermally assisted transport. The currents observed in bipolar switching of oxides are often in the low mA range, at least in the some ten μA range. In the case of confined electrical conduction, this may easily lead to a local temperature rise by some 100 K and, therefore, a considerable acceleration of the drift. A third possibility is given by the extremely high fields at the switching interface. As described by Equation 4 in Section 3.2, the mobility becomes field-accelerated at very high fields. In the OFF-state of VCM cells, fields in the order of 1–2 V nm⁻¹ may occur which may decrease the activation energy by 0.2–0.4 eV and lead, therefore, to an acceleration by some orders of magnitude.^[139] In the case of the oxide dual-layer concept reported by Meyer et al.,^[135] a thermal activation of the oxygen vacancy transport has been ruled out due to the local currents involved. However, the authors have shown that the electric field in the tunnel oxide and the adjacent Debye length of the conductive metal oxide can be up to 10 MV cm⁻¹ during SET and RESET. The field enhanced mobility of oxygen vacancies may increase by several orders of magnitude compared to the low field mobility at room temperature. A comprehensive analysis of the field-

accelerated ion mobility has shown that a combined acceleration by local high temperatures and very high electric fields may lead to an increase in the ion mobility of > 12 orders of magnitude.^[140] The fields required for a sufficient acceleration, however, often approach or exceed the breakdown fields of the materials, as long as the local field at the ion position is not enhanced in comparison to the average field.^[141] As mentioned in Section 2.1, all RRAM types need to overcome the voltage-time dilemma of ultra-fast write processes and a very long retention under read voltage stress. At present, it is not yet possible to draw a final conclusion on the impact of the field-accelerated ion mobilities on overcoming this dilemma.

In PCM cells, for example, the voltage-time dilemma is overcome by thermally triggered phase change and the temperature dependence of the viscosity of the amorphous phase based on the non-conventional Vogel–Fulcher mechanism. Drift-diffusion processes, even with the non-linearities generated by strong concentration changes, are far away from being able to overcome the voltage-time dilemma. Obviously, there must be a mechanism or a combination of mechanisms involved which give rise to a sufficiently high non-linearity. Two possibilities have been mentioned above, the thermal activation and the field enhancement of the oxygen vacancy transport. Furthermore, it has been proposed that a lattice strain caused by a high oxygen vacancy concentration may affect the diffusion coefficient.^[125]

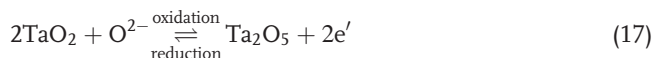
Another possibility is the formation of another phase. Non-stoichiometries in transition metal oxides only show a limited thermodynamic stability range. If the concentration of defects such as oxygen vacancies exceeds these limits the phase becomes instable and another, more stable phase may be formed. This process may involve meta-stable phases as well. For example, Szot et al. have shown the electromigration induced formation of new phases such as Ruddlesden–Popper type phases in BaTiO₃ after passing a charge of 5 C through a polycrystalline cell at 770 K.^[142] In the context of resistive switching of oxides, a deliberate strategy has been set up recently to use the knowledge about the stability of transition metal oxides with different stoichiometry in order to realize VCM type memories. Odagawa et al. reported about bipolar switching in magnetite thin films.^[143] Magnetite, Fe₃O₄, has the highest conductivity among the Fe oxides. After electroformation, resistive switching has been possible by bipolar voltage pulses. The switching is interpreted in terms of oxygen ion migration and their accumulation in a region near the anode. Obviously, the redox reaction



lead to the formation of a $\gamma\text{-Fe}_2\text{O}_3$ layer which fully covers the anode. Because of the significantly lower conductivity of $\gamma\text{-Fe}_2\text{O}_3$ the cell is switching into the OFF state at this polarity. The formation and removal of the $\gamma\text{-Fe}_2\text{O}_3$ phase was confirmed by Raman spectroscopy.

Utilizing the same concept, Wei et al.^[144] have introduced a TaO_x based memory cell which they have fabricated in 180 nm technology. The bulk of the oxide film consists of a TaO₂ which shows a smaller band gap and a significantly higher conductivity than the fully oxidized Ta₂O₅ phase. At the active electrode of a Pt/TaO_x/Pt cell the bipolar switching pulse moves oxygen ions to or from the layer intimately attached to the Pt metal and,

therefore, changes the stoichiometry from an oxidized Ta₂O₅ phase (OFF-state) to the reduced TaO₂ phase (ON-state) according to the redox reaction



Using hard-XPS, the authors found evidence for the valence change which accompanies the phase transformation. Switching times in the order of 10 ns, endurance of $> 10^9$ cycles, and extrapolated retention times of > 10 yrs are reported.

Because of the threshold character of the process, the electrochemically driven phase transformation can easily overcome the voltage-time dilemma. However, many questions still need to be addressed. First of all, the formation of a phase on the nanoscale needs to be experimentally investigated for VCM systems in general, including those aspects discussed in Sections 4.4 and 4.5. Second, the speed of nucleation of the phase and material parameters affecting the nucleation rate need to be studied. Third, the critical size of the new phase and the corresponding overpotential for generating it must be studied. In particular this last point is essential in order to estimate the scalability potential.

5. Thermochemical Systems

5.1. General Features

There are several types of electrically controlled resistive switching which are primarily based on thermal effects and show a unipolar characteristic. The most prominent type is the phase change effect utilized for PCM devices which is not covered by this review. Instead, we will describe a thermochemical switching which is frequently observed in transition metal oxides. The most prominent candidate material out of many is NiO, first reported about in the 1960s.^[145] Figure 33 shows a typical log I – V

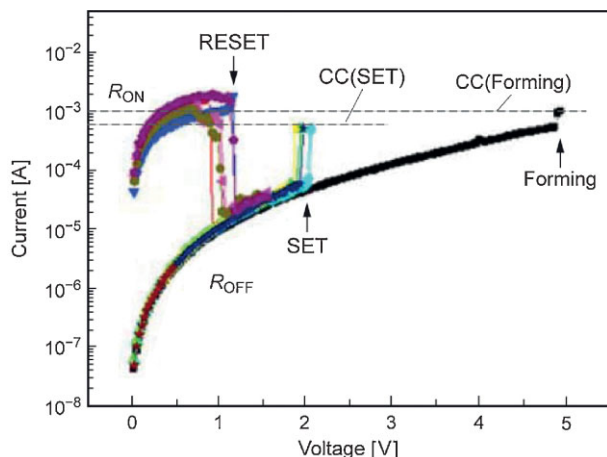


Figure 33. Unipolar current–voltage characteristic of a Pt/NiO/Pt stack with a NiO film thickness of 50 nm and $100\ \mu\text{m} \times 100\ \mu\text{m}$ Pt top electrodes. Reproduced with permission from [146]. Copyright 2007, Wiley. For a reason described below, the top electrodes were fabricated with a thickness of only 5 nm. The “Forming”, “SET”, and “RESET” processes are indicated. During the forming cycle, a compliance current of 1 mA has been set, while for all subsequent SET processes it is reduced to 0.5 mA.

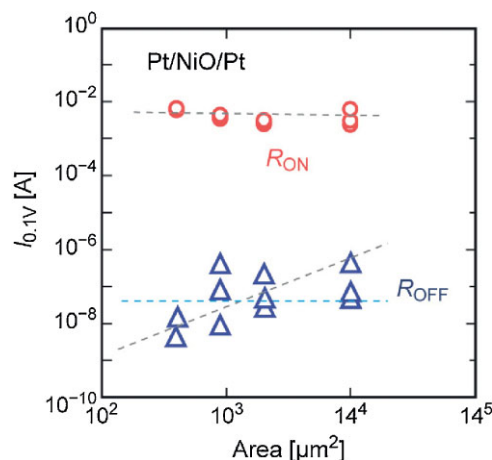


Figure 34. Top electrode area dependence of the current at $V = 0.1$ V for a Pt/NiO/Pt stack with a NiO film thickness of 50 nm in the OFF and in the ON state. Reproduced with permission from [151]. Copyright 2007, AIP. The dashed lines are guides to the eyes. It should be noted that the OFF resistance is also reported as area independent. Reproduced with permission from [154]. Copyright 2004, IEEE.

characteristic of a NiO thin film with Pt electrodes.^[146] Within the first sweep starting at 0 V, the MIM system exhibits a sudden current increase at a forming voltage of approx. 5 V. The current increase is limited by a compliance at 1 mA. Subsequently, the system can be RESET by releasing the current compliance and, again, SET at a voltage much below the forming voltage utilizing a suitable current compliance.

The virgin resistance^[147] and sometimes also the OFF resistance is found to be approximately proportional to the electrode area suggesting a uniform current density in the oxide over the cross-section of the electrode area for NiO-based MIM systems,^[148–150] as well as for other TMO films such as CoO^[151] and Fe₂O₃.^[152] The ON resistance is considered to be caused by filamentary conduction paths in all reports in the literature. However, there is some debate if the ON state of a MIM cell is caused by a *single* filament as indicated by an electrode area independent ON resistance (Fig. 34)^[148,150,151,152] or by *multiple* filamentary current paths, which are randomly distributed.^[149,154]

The ON and OFF resistances are only slightly temperature dependent as shown for Pt/NiO/Pt systems in the temperature range from 10 K to 300 K^[155] (Fig. 35). The temperature dependence of the ON resistance typically shows a weak metallic behavior for which a relative resistance ratio (RRR) defined as $R(300\text{ K})/R(5\text{ K})$ of 1.6 has been found.^[156] This value is well below typical values of pure metals and indicates a defective, impure metallic conducting phase. Within the operating temperature regime of electronic devices, the OFF resistance shows the signature of a semiconductor with a thermal activation of approx. 0.1 to 0.15 eV^[150,151,155,156] and a magnitude which is similar to the resistance of the virgin, unformed NiO thin film. However, below a specific threshold temperature, which is in the cryogenic regime, the temperature dependence of the OFF resistance changes drastically and becomes weakly metallic.^[155]

Based on these phenomenological dependencies it is assumed that this type of unipolar resistive switching is triggered by a filamentary thermal breakdown of the oxide leading to a con-

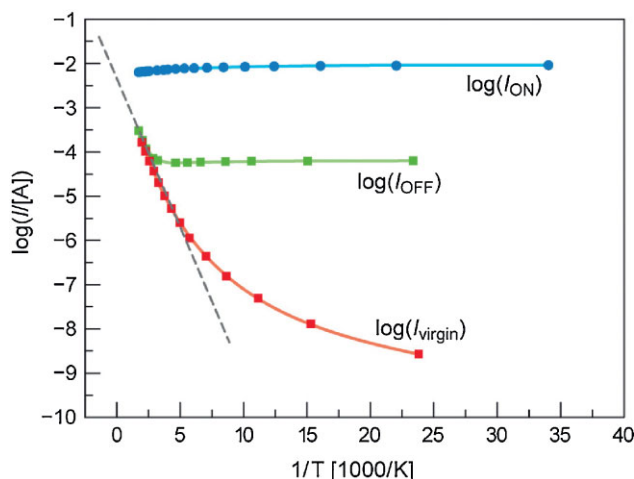


Figure 35. Temperature dependence of the conductivity of Pt/NiO (300 nm)/Pt thin films in the virgin state (red), the OFF state (green), and the ON state (blue). Data are redrawn and assembled from [155]. Copyright 2007, AIP.

duction channel between the electrodes. Due to the compliance current during the SET process, only a weak conductive filament with a controlled resistance is formed. During the RESET transition the current compliance is held inactive, so that this conductive filament is disrupted thermally again – similar to blowing a traditional household fuse, however, on a nanoscale. For this reason, this switching mechanism is sometimes referred to as the fuse – antifuse type. In the next section, we will discuss the current understanding of the microscopic mechanism.

5.2. Phenomenological Description of the Switching Mechanism

Above a critical electrical field strength all materials suffer from an electrical breakdown. Typically this breakdown is induced by a thermal runaway.^[36] By applying a field E , the residual conductivity σ of the insulating material, such as transition metal oxides, leads to the local generation of Joule heat. This energy is balanced by a temperature increase and a heat conduction according to

$$\sigma E^2 = C_V \frac{\partial T}{\partial t} - \text{div}(\kappa \text{grad} T) \quad (18)$$

where C_V is the specific heat per unit volume and κ is the thermal conductivity. The runaway process is caused by the exponential temperature dependence of the conductivity which is found in all insulating and semiconducting materials,

$$\sigma \sim \exp\left(-\frac{W_A}{kT}\right) \quad (19)$$

In this expression, W_A denotes the activation energy of the conductivity, usually caused by the temperature dependence of

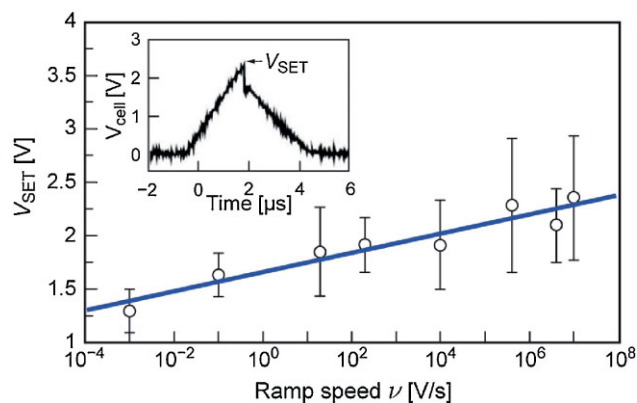


Figure 36. SET voltage, V_{SET} , as a function of the sweep rate, ν , for Pt/NiO/Pt cells with a NiO thickness of 160 nm. Dots show experimental data, the blue line shows an empirical fitting. Reproduced with permission from [157]. Copyright 2008, IEEE.

the carrier concentration. In the case of short pulses which are used to operate RRAM devices, the conduction term in Equation 18 can be neglected, i.e., the situation can be treated quasi-adiabatically.

As for the other resistive switching classes ECM and VCM, the SET process of TCM cells can be conducted by a voltage sweep or a pulse. The dependence of the voltage V_{SET} on the ramp rate or the pulse width shows a logarithmic behavior as discussed in detail for ECM systems (Fig. 36).^[157]

The localized thermal runaway seems first to lead to an electronic ON state which does not result in a permanent change of the resistance and, hence, is described as a *threshold* switching. If the stimulation continues, this transient ON state leads to a local redox reaction, obviously because of oxygen drift out of the high temperature region due to the energetically favored lower valence states of the transition metal oxide and, eventually, to a structural modification. Only this process establishes the permanent ON state, i.e., the *memory* switching. This interplay of threshold and memory switching is determined by the thermal properties of the cell, e.g., the heat conduction of the electrodes,^[157,158] and it shows some interesting similarities to PCM cells.

The driving force which favors lower valence states at higher temperatures is generic to all stable oxides because of the negative free energy of formation, as can be seen from the Ellingham diagram (Fig. 37). This is the most important reason why TCM effects are found in all transition metal oxides, and even beyond e. g. in GaO_x .^[160] Starting at a given (local) oxygen activity at room temperature for any of the higher valent oxides, a temperature rise of some hundred K (starting at 300 K and moving to the right in Fig. 37) will always lead to a stable oxide with a lower valence or to the corresponding metal.

Lateral Pt/CuO/Pt cells have been used to study the valence of the metal in the conductive filament. Because of the minute amount, this has been a difficult task. Figure 38a shows the SEM image of the cell clearly displaying the conductive path between the two electrodes. In Figure 38b, a PEEM image is a divide image of the XAS CuO absorption edge (930.3 eV) and the XAS Cu_2O absorption edge (932.6 eV) recorded at the Spring-8/JASRI facility.^[161] The latter figure shows a percolative line of a lower Cu

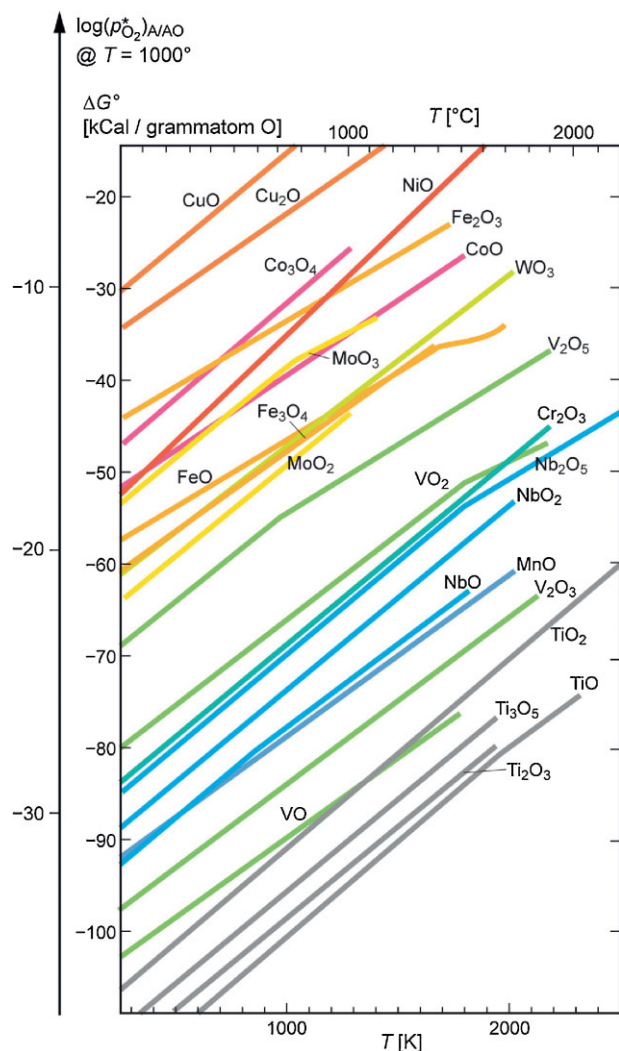


Figure 37. Temperature dependence of the free energy of formation of various transition metal oxides. The energy axis is also shown in equivalent (formal) equilibrium oxygen partial pressures. The color code shows phases of identical cations (redrawn from [159]).

valence along the conductive filament, supporting the hypothesis described above.

A controversy still exists in the literature, about the localization of the switching. The reports mentioned so far as well as several others are in favor of a single filament switching model. However, there are reports which clearly indicate that a multifilament switching may be possible as well. Son and Shin have used a Hg drop top electrode to switch a NiO film. After switching into the ON and OFF state, respectively, they removed the Hg drop and investigated the surface by LC-AFM (Fig. 39).^[162] In this experiment obviously the density of the conducting filaments change drastically between the ON and OFF state. It is interesting to note that the filaments in the OFF state (and partially also in the ON state) are mainly located at the grain boundaries of the NiO film.

The temperature dependence of the resistance in the ON-state of Pt/NiO/Pt systems shows a slightly positive coefficient indicative of metallic conduction. Russo et al. performed an electrothermal

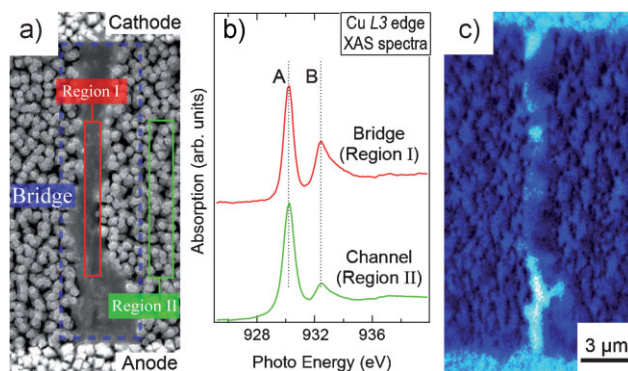


Figure 38. a) SEM image of the planar-type Pt/CuO/Pt resistance-switching device. Top and bottom regions in the image correspond to the anode and cathode parts, respectively. The central lighting-shaped structure is the bridge structure formed in the CuO channel. b) XAS spectra of Cu L3 absorption edge for bridge structure (Region I) and CuO channel (Region II) structures. The reduction component at 932.6 eV (B) is enhanced at the bridge structure. c) Division of two PEEM images taken in the same region as the SEM image and obtained with a photon energy of 930.3 eV (CuO derived component A) and 932.6 eV (reduction component B), respectively. The bright regions in the bridge structure correspond to the reduced region of the CuO channel. Reprinted with permission from H. Kumigashira, [161].

simulation of cylindric metal filaments in a NiO matrix, assuming a single filament switching, to match their experimental RESET curves.^[163] Figure 40 shows some of their results. The dissolution of the filament was phenomenologically described by a thermally activated process of the filament surface. As Joule heating increases (A), a high dissolution zone develops in the middle of the filament leading to a hot spot (B). In a self-accelerating process, this locally enhances the electric field and current density, hence the Joule dissipation and temperature increase, which in turn enhances the velocity of dissolution (C) until rupture (D).

Although the simulation describes the experimental data surprisingly well, in particular the *I-V* of the RESET process, a number of questions concerning the microscopic physics remain. Despite the metal like resistivity confirmed for the conductive filaments (as, for instance, for the ON-state of filaments in SrTiO₃ thin films, see Sec. 4.5), the crystallographic phase is not clear yet. This question is linked to the physical meaning of the thermal dissolution in the simulation and the resulting critical temperature, which is much too low (approx. 530 K) to indicate the melting of a Ni metal filament, the inherent switching times, and the reliability (data retention, fatigue cycles, etc.). Chang et al. reported electro-thermal simulations of TCM cells in which they emphasized the influence of the metal electrodes in the process.^[158] It must be mentioned that alternative models concerning the location of the unipolar resistive switching in transition metal oxides have been reported in the literature. A prominent alternative model has been presented by Inoue et al.^[164] in which the switching is supposed to take place at the interface of conducting filaments with the electrodes on both sides. The percolative nature of the process has been studied by Chae et al. and worked into a detailed model which includes the reversible dynamic processes during the TCM type resistive switching.^[165]

A critical parameter for this unipolar switching effect seems to be the value of the compliance current. In fact, it has recently been

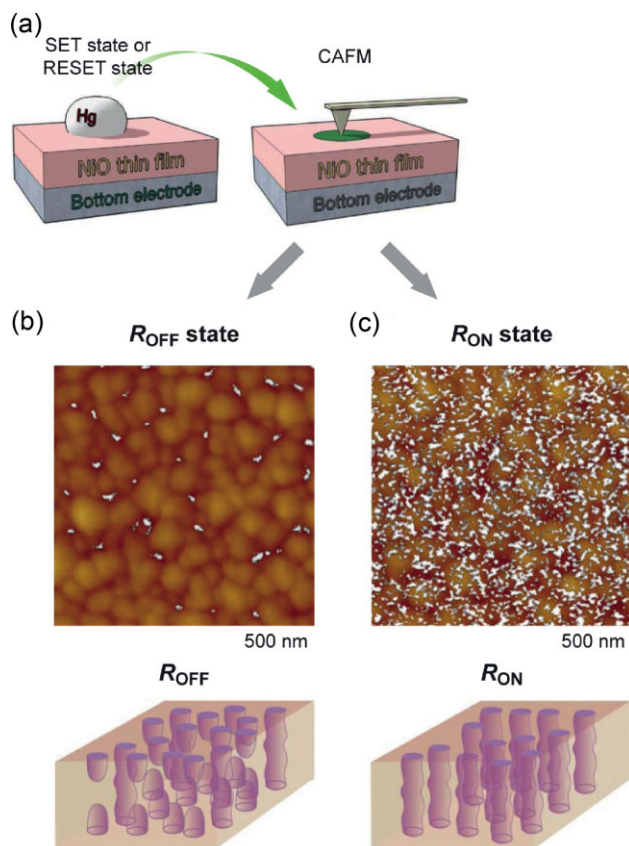


Figure 39. Mapping of the conductivity of the ON and OFF state of a NiO thin film. a) sketch of a removable Hg top electrode on a NiO thin film for resistive switching studies and of the LC-AFM study after switching, b) and c) LC-AFM image of the NiO film recorded at 0.1 V after 100 switching cycles and model of the multifilament scenario in which the filaments are disconnected (OFF state (b)) or connected (ON state (c)). Reproduced with permission from [162]. Copyright 2008, AIP. The images represent overlays of the topography and the local conductivity where the white areas indicate regions of high conductivity.

shown that a TiO_2 thin film exhibits bipolar switching, and, on setting the compliance current to a larger value, can be turned into a unipolar switching characteristics.^[126]

These are clear indications that the voltage-time dilemma for TCM under the memory switching scenario may also be overcome by the local formation of different phases (on the nanoscale). Again, many details need to be studied such as the stability, the critical size of the nucleus, the parameters affecting the phase formation rate. The answer to these questions will provide a more stable base for all essential TCM-based RRAM specifications and the reliability issues.

6. Scaling Potential and Alternative Architectures

There have been several projects aiming at the integration of redox-based resistive switching cells into the conventional active-matrix CMOS memory architecture in order to realize a high-density non-volatile RRAM.

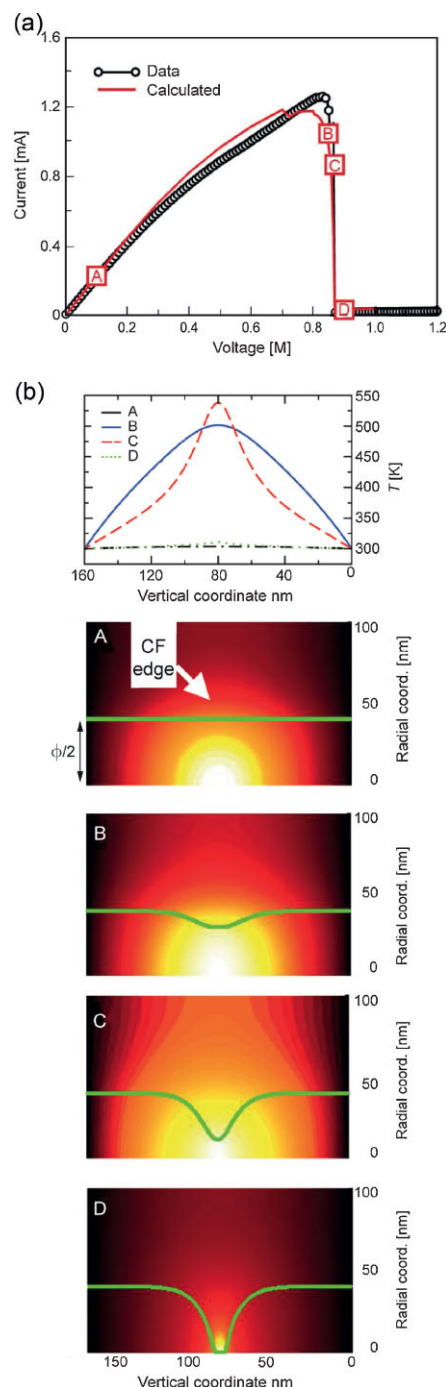


Figure 40. a) Measured and calculated I - V curve during RESET of a 160 nm thick NiO film on n -Si and with Au top electrodes. Bias points A, B, C and D, corresponding to simulation results in the figure below. b) Simulation results for thermal dissolution of the conductive filament CF. The dissolution is faster at the hot spot (the middle of the filament in the figure). The formation of a bottleneck further enhances Joule heating due to current crowding, resulting in a self-acceleration of RESET. Also shown (right) is the temperature profile along the cylindrical axis in the filament, for the four bias points. ϕ denotes the filament diameter (between 40 and 80 nm). Reproduced with permission from [163]. Copyright 2007, IEEE.

The state of the art with respect to integration of ECM cells into RRAM was demonstrated by a R&D team at Qimonda AG^[166–168] who developed Ag/GeS(e)_x/W cells integrated into a 90 nm CMOS technology in an active matrix concept with a select transistor (so-called 1T1R memory). Figure 41 illustrates a sketch of the cross-section and the cell schematic of a one transistor/one ECM junction memory cell. In this figure, two ECM cells are displayed, each connected to a common bit line (BL). The connection of the ECM cells through the adjacent select transistor to the BL is accomplished by activation of either wordline WL(0) or WL(1), respectively. The Ag metal electrodes of the ECM cells form a common plate line (PL). An integration density of 2 Mbit as well as a multi-level 2 bit/cell operation have been achieved. Switching times <50 ns, endurance of >10⁷ write cycles, and an extrapolated data retention >10 years at 70 °C have been shown. In another paper on integrated cells, Kozicki et al. reported an endurance of well above 10¹⁰ write cycles for a single memory cell.^[25]

Already few years ago, NiO-based TCM cells have been fully integrated with a 180 nm CMOS technology as reported by a team of Samsung Electronics.^[147,153] It has been demonstrated that the materials involved are highly compatible with the conventional CMOS process and that no other dedicated process is required.

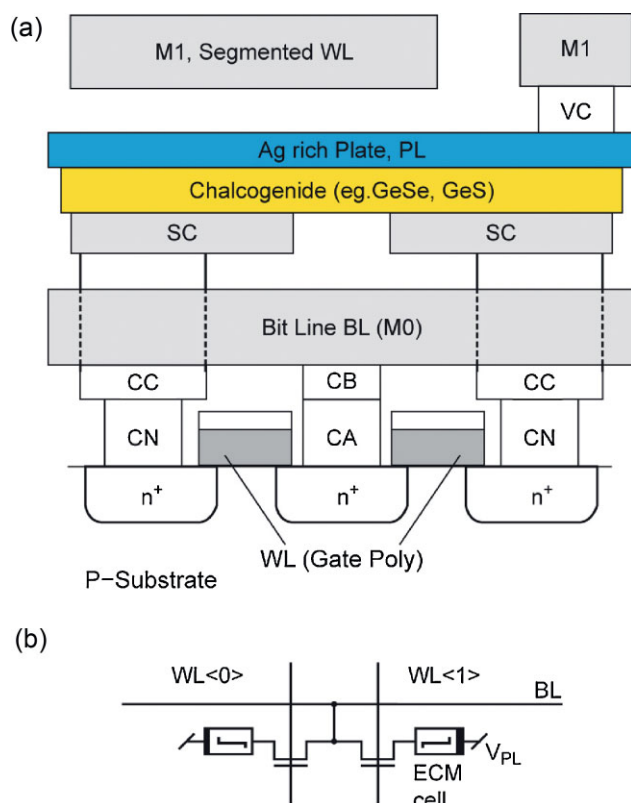


Figure 41. a) Cross-section and b) schematic of an integrated ECM memory cell. WL pitch = BL pitch = $2F = 180$ nm. M1: hierarchical metalization level; VC: Via Contact; SC: Storage Contact; CC: Cell Contact; CB: Contact Bitline; CA: Contact Array device; CN: Contact Node; WL: Word line; BL: Bit line; PL: plate line; F denotes the minimum feature size of the process technology, here: 90 nm. Reprinted with permission from [168]. Copyright 2007, IEEE.

The integrated 1T1R cells showed a write endurance of 10⁶ cycles, fast switching, and a R_{OFF}/R_{ON} ratio of more than 10 with excellent temperature stability up to 300 °C. Recently, also TaO_x VCM cells have been integrated in a 180 nm CMOS technology.^[144] The 1T1R memory cells showed stable pulse switching with an endurance of 10⁹ cycles and an extrapolated data retention exceeding 10 years at 85 °C.

The passive matrix crossbar concept is even more promising than the conventional active matrix CMOS approach because several reasons. One reason is the inherently high density which allows for $4F^2$ cell, where F is the minimum feature size of the fabrication technology. A second reason is the chance to further shrink the size of the passive matrix, without the need to do the same for the MOSFETs.^[169] The third reason lies in the fact that a potential 3D stacking is easier and more effective for passive crossbars than for active CMOS circuits.^[170] A drawback of passive crossbars is given by the fact that the memory elements cannot be electrically isolated while neighbor cells are addressed, i.e., the parasitic-path-problem occurs as mentioned above. This problem can be solved by serial elements with a specific (high) non-linearity at each resistively switching cell, depending on the resistive properties and the array size.^[2,171] Preferably, this nonlinear serial element is compatible with the transition metal oxide used for the resistive switching.^[172]

The passive crossbar concept opens opportunities which go far beyond pure memory devices. In a pioneering work by Heath et al. in 1998,^[173] it has been proposed to build an entire computer based on a nanowire crossbar with resistively switching cells (considering complex organic molecules at that time). These crossbars would be the reconfigurable look-up tables of a free programmable gate array (FPGA) circuit which is assembled into a computer by a so-called fat global interconnect tree. The advantages of such a computer architecture over the traditional von Neumann architecture are established by an inherent defect tolerance which can be built into the system, the local fusion of logic and memory which increases the energy efficiency because mainly local interconnect lines are charged and discharged, and by a largely reduced overhead of a control-and-glue logic. There have been various approaches to these alternative architectures.^[174–176] An efficient solution of the problem of decoupling the shrinking feature size of a passive crossbar matrix with respect to a more crude CMOS circuits underneath was proposed by Likharev.^[169] His concept introduces a tilting angle between the crossbar matrix and the array of CMOS gates which puts the contact points between the passive crossbar and the CMOS array on a larger registry (Fig. 42).

As briefly mentioned in Section 2.1, multilevel resistance switching has been observed for various redox-based RRAM systems (see, e.g.,^[13,15,17,116]). For instance, for Ag/Ag-Ge-Se/Pt cells as well as for Cu/SiO₂/Pt cells in vias of Si₃N₄ layers, the tuning range of the write current has been studied. As mentioned in Section 3.1, the ON resistance after completion of the SET process is determined by the write current through a compliance limit which is established for positive voltage polarity at the active electrode. This limits the strength of the filament and, in particular, the strength of the neck between the Cu (Ag) filament and the Cu (Ag) electrode. For example, a write current of 25 μ A has been used in Figure 4. By setting the write current appropriately, the ON resistance can be set over many orders

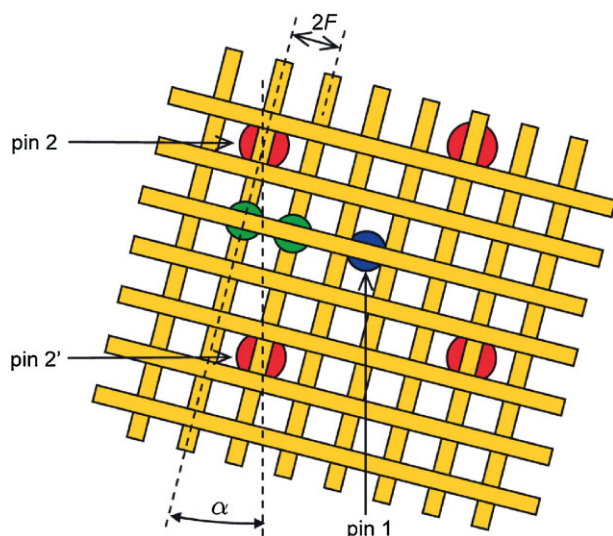


Figure 42. a) Schematic top view of the contact points of an array of CMOS devices (red dots and blue dot in the center) and a passive crossbar matrix (yellow stripes) on top, tilted by an angle α . Reproduced with permission from [176]. Copyright 2005, IOP. Because of the tilting, individual nodes of the passive crossbar (e.g., the green dots) can be addressed (by pin 1/pin 2 and pin 1'/pin 2', respectively) although the registry of the CMOS contacts is much coarser.

of magnitude as shown for the two different ECM systems (Fig. 43).

The arrow in this figure indicates the Landauer limit which marks the galvanic contact of one atom. Obviously, stable ON states can be achieved well below this limit representing a situation, in which a controlled tunneling gap is left in the ON state. Because of the large dynamic range of ON resistances over more than six orders of magnitude, these ECM systems are particularly suitable for multilevel applications. In fact, the memristive characteristics of most redox-based systems are constituted by the fact that the resistance value written into the cell is an analogue value. This opens the avenue to a completely new functionality, namely neuromorphic networks. In such a concept, the synapses of the artificial neurons of these networks would be realized by redox-based memristive cell utilizing the ability to program and store analogue resistance values which then represent the synaptic weight.^[177,178] However, despite the prospects, many fundamental aspects of this new concept remain to be investigated before any real application is in sight, such as the retention time of the analogue resistance values, the impact of statistical deviations between devices, etc.

In addition to novel functionalities, redox-based resistive switches also show a high scalability potential in comparison to other memory technologies. The scalability potential of integrated ECM cells has been approached by fabricating an inert Si_3N_4 film on top of a W bottom electrode and etching vias of defined diameters into the Si_3N_4 film.^[166] The vias were filled with W, followed by layers of GeSe and Ag top electrodes. Successful switching operation has been shown for all via diameters down to 20 nm, which have been the smallest of this study. This clearly demonstrates that the scalability limit of this type of ECM cells must be below 20 nm.

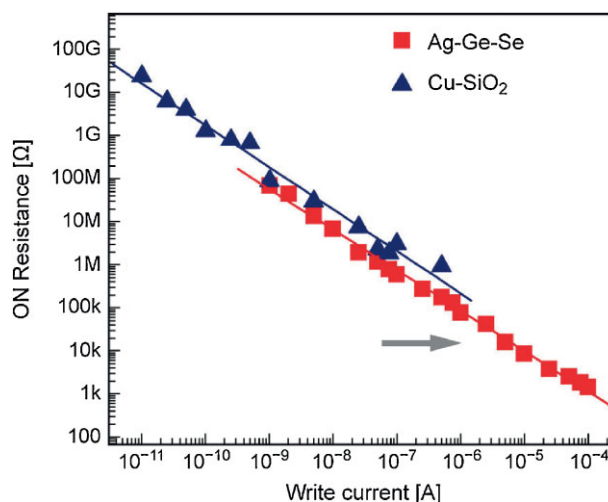


Figure 43. Tuning the ON resistance by write current variation in ECM cells. Reproduced with permission from [21,34]. Copyright 2007, IEEE and 2008, AIP respectively. The extremely broad range of ON resistances is particularly interesting for multi-level RRAM cells. The Landauer limit (arrow) is approx at 12 kΩ.

In search of the scalability limit of the actual contact of ECM cells, Terabe et al. used a passive cross point between a Ag_2S coated Ag wire and a Pt wire including a 1-nm tunneling gap between the surface of the Ag_2S and the Pt.^[27] The crossbar structure has been fabricated by electron-beam lithography, film deposition, and lift-off. On top of the Ag_2S coated Ag wire, the 1-nm tunneling has been realized by a 1-nm thick Ag layer before the Pt wire was deposited. This configuration represents the cell in an initial ON state with a resistance of a few kΩ. By applying an electroforming step through a positive RESET voltage to the Pt electrode, the 1-nm thick Ag layer is electrochemically dissolved and incorporated into the Ag_2S layer, leaving a 1-nm-thick vacuum gap between the Pt electrode and the Ag_2S electrolyte. Upon applying a SET voltage, the nanogap acted as an electron injection layer due to tunneling which led to the formation of an Ag protrusion which closed the gap and contacted the Pt electrode. It has been possible to control the SET process to the level of quantized electron transmission channels described by discrete Landauer conductance levels, i.e., multiples of $2e^2/h$, which demonstrates, in principle, the extreme scalability potential of ECM cells. A Landauer conductance of $2e^2/h$ represents the galvanic contact through just one Ag atom.

In the case of VCM cells, the ultimate scaling potential in the case of the filamentary mechanism seems to be given by the conduction zone around a switching dislocation intersecting with the electrode.^[103] For TCM cells, the scaling limits have not yet been addressed in detailed studies.

7. Summary

This review describes three different types of resistive switching memory, short RRAM, in which redox processes and ionic motion on the nanoscale play the key role. These three types rely on an electrochemical metallization mechanism (ECM), a valence change mechanism (VCM), and a thermochemical mechanism (TCM).

The report focuses on the current understanding of the microscopic mechanism of the three switching types and addresses the crucial question of the voltage-time dilemma which needs to be overcome in order to utilize the concept in universal non-volatile memories. It shows that typically redox-based phase transformations seem to be involved which builds up an interesting link to the purely thermal phase change memories (PCM).

In order to explore the potential of these types of RRAM further and to exploit their potential to the limits, a considerable research effort is still needed with respect to a deeper understanding of the microscopic mechanism of the switching. The process and material optimization, the effects limiting the reliability, numerous aspects of fabrication technology, as well as the guide lines of scaling need to be addressed. In particular, the cross-effects between the chemical, thermal, and electronic phenomena involved in the resistive switching mechanisms discussed in this review are only elucidated to a very small degree. However, the steadily increasing number of excellent publications by groups all over the world is a promising sign that this challenging mission has been acknowledged and adopted.

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